BSNet: Dynamic Hybrid Gradient Convolution Based Boundary-Sensitive Network for Remote Sensing Image Segmentation

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***Abstract*— Boundary information is essential for the semantic segmentation of remote sensing images. However, most existing methods were designed to establish strong contextual information while losing detailed information, making it challenging to extract and recover boundaries accurately. In this article, a boundary-sensitive network (BSNet) is proposed to address this problem via dynamic hybrid gradient convolution (DHGC) and coordinate sensitive attention (CSA). Specifically, in the feature extraction stage, we propose DHGC to replace vanilla convolution (VC), which adaptively aggregates one VC kernel and two gradient convolution kernels (GCKs) into a new operator to enhance boundary information extraction. The GCKs are proposed to explicitly encode boundary information, which is inspired by traditional Sobel operators. In the feature recovery stage, the CSA is introduced. This module is used to reconstruct the sharp and detailed segmentation results by adaptively modeling the boundary information and long-range dependencies in the low-level features as the assistance of high-level features. Note that DHGC and CSA are plug-and-play modules. We evaluate the proposed BSNet on three public datasets: the ISPRS 2-D semantic labeling Vaihingen, the Potsdam benchmark, and the iSAID dataset. The experimental results indicate that BSNet is a highly effective architecture that produces sharper predictions around object boundaries and significantly improves the segmentation accuracy. Our method demonstrates superior performance on the Vaihingen, the Potsdam benchmark, and the iSAID dataset in terms of the mean F_1, with improvements of 4.6%, 2.3%, and 2.4% over strong baselines, respectively. The code and models will be made publicly available.**

***Index Terms*— Attention module, boundary information enhancement, remote sensing, semantic segmentation.**

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I. Introduction

With the rapid development of remote sensing technology, an increasing number of remote sensing images with high resolution (e.g., 5–10 cm) can be obtained. In these images, small objects, such as cars and trees, can be clearly observed, which makes pixel-level semantic segmentation possible. Image semantic segmentation allocates pixelwise semantic labels in an image, segmenting the image into several regions based on semantic units. Semantic segmentation of remote sensing images has a wide range of applications, including environmental monitoring [1], postdisaster reconstruction [2], agriculture [3], forestry [4], and urban planning [5], [6]. Traditional image semantic segmentation methods [7]–[10] manually design feature classifiers but do not achieve satisfactory performance due to their limited feature extraction and data fitting abilities. These shortcomings have prompted researchers to search for more stable and efficient methods.

In recent years, many image semantic segmentation algorithms have emerged. CNN-based methods significantly outperform other methods and have gradually become a research hotspot. The fully convolutional network (FCN) [11] has achieved a massive breakthrough in semantic segmentation. It replaces the fully connected layer with a convolutional layer, allowing image segmentation at any resolution to achieve end-to-end training and prediction. In particular, the encoder–decoder architecture [11], [12] has proven to be highly effective in end-to-end semantic segmentation tasks. Despite this success, continuous downsampling in the encoder network results in the loss of high-frequency detailed information, which suggests that the segmentation near the boundaries is usually unsatisfactory.

To alleviate the above problems, a variety of strategies [13]–[19] have been proposed that can be roughly classified into the following three families: skip connections, multiscale contexts, and additional boundary branches. Skip connections [20]–[22] embed low-level features that contain boundary information into high-level features. Alternatively, multiscale contexts expand the receptive field under the premise of maintaining high resolution to generate semantic feature maps with strong context relationships [23]–[26], such as the pyramid pooling module (PPM) [24] and atrous spatial pyramid pool (ASPP) [25]. Additional boundary branches [14],

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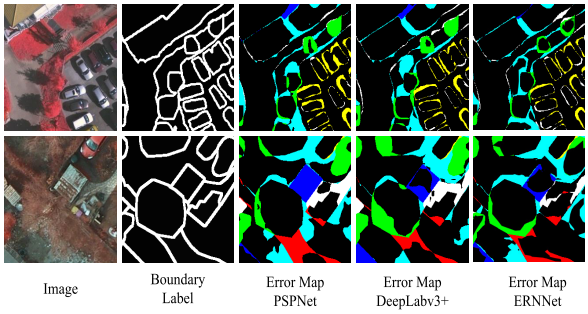


Fig. 1. Problem illustration. The black area in the error map represents the correct prediction. It can be observed that the prediction errors of segmentation are mainly located close to the object boundaries. The first and second rows show examples from the Vaihingen and Potsdam datasets, respectively.

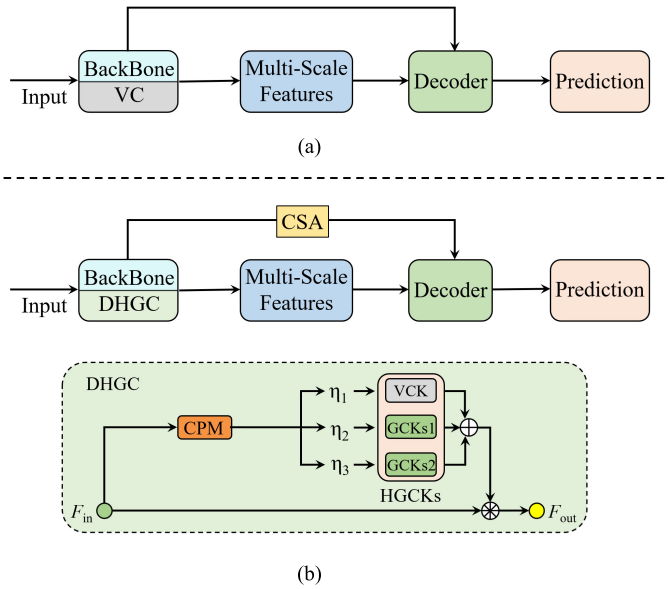


Fig. 2. Illustration of the segmentation pipeline. (a) In previous work, VC was used to extract features and directly introduce low-level features into high-level features to recover boundaries. It is not sufficient for remote sensing images with complex boundaries. (b) In our method (BSNet), in the encoder network, DHGC enhances boundary information extraction through the adaptive aggregation of one VCK and two GCKs. The GCKs explicitly encode the boundary information. The adaptive coefficients (η_i) are generated by the CPM. In the decoder network, the boundary information and long-range dependencies of low-level features are adaptively mined by the CSA to assist high-level features in recovering clearer and sharper boundaries.

[15], [27]–[29] are also a popular method. For example, Liu *et al.* [15] designed multiple weighted edge supervisions and EALoss to extract boundary features. Li *et al.* [14] proposed a semantic boundary awareness network (SBANet), which emphasizes useful boundary information by boundary attention modules and adaptive weights in multitask learning.

However, these methods still cannot achieve satisfactory results. As shown in Fig. 1, the prediction errors near the object boundaries of remote sensing images are still terrible. We take two main issues into consideration.

First, the boundary information is difficult to be explicitly encoded. As shown in Fig. 2(a), existing methods use vanilla convolution (VC) to extract dense features. Although VC has an advantage of discriminative semantic representation, it does not restrict explicit gradient encoding, resulting in poor gradient information extraction and, consequently, the degradation of boundary extraction [30]. In addition, remote sensing images often have complicated boundaries due to numerous small fragments, ubiquitous shadows, occlusions, diverse backgrounds, and complex textures. Thus, the methods applied to remote sensing images have stringent and urgent requirements for sufficient boundary information extraction. Moreover, how to enhance boundary information extraction adaptively through explicit encoding while ensuring semantic information extraction is also a challenge.

Second, high-level features in the decoder are of unclear boundaries. High-level features are good at category prediction, while low-level features have rich spatial details. Remote sensing images usually have large scenes and complex spatial composition due to their imaging conditions, which increases the necessity of introducing high-resolution spatial information from low-level features to recover boundary details. Existing methods directly introduce low-level features into high-level features of the decoder network. However, input images with different boundary complexities have different requirements for boundary information components from low-level features. Therefore, it is unreasonable to directly introduce low-level features into high-level features. How to adaptively model the boundary information from low-level features is very crucial. Furthermore, the spatial information in low-level features has local dependencies but fails to obtain long-range dependencies. Long-range dependencies are essential for segmentation tasks, let alone boundary prediction.

In response to the above issues, this article proposes three key strategies: the gradient convolution kernels (GCKs), the dynamic hybrid gradient convolution (DHGC), and the coordinate sensitive attention (CSA), as demonstrated in Fig. 2(b).

For the first issue, it is well known that boundary information explicitly corresponds to drastic gradient changes. Early classical methods use the first- or second-order derivatives (e.g., Sobel [31], Prewitt [32], Laplacian of Gaussian (LoG), and Canny [33]) for basic edge detection. Later learning-based methods [34], [35] further predict more precise boundaries by utilizing various gradients' information, such as color, brightness, texture, and depth [36]–[39]. Inspired by the above wisdom, we believe that the joint modeling of the operator and CNN-based method can build a new learnable convolution that can easily extract boundary information of numerous input images. In this article, we correspondingly proposed the GCKs. GCKs are composed of a pair of kernels, denoted as GCKs1 and GCKs2.

At the same time, we still have to consider ensuring that both semantic information and boundary information are fully extracted. It occurred to us that the traditional hybrid spatial enhancement method [40] can combine a variety of complementary methods for image enhancement to achieve the effect that cannot be achieved by one method. Inspired by this wisdom, we combine VC kernel (VCK) and two GCKs into parallel hybrid GCKs (HGCKs) to extract both semantic and boundary information. The weight between VCK and GCKs is adjusted according to the image content, which is determined via the coefficient prediction module (CPM). This is implemented as the DHGC, which aggregates one VCK

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and two GCKs into a new efficient operator to enhance the boundary information in the encoder network.

For the second issue, we develop the CSA module based on the popular CA module [41]. Instead of introducing low-level features directly to high-level features, the CSA adaptively extracts the boundary information of low-level features and further mines its long-range dependencies to help high-level features recover more accurate boundaries. Inherited from the wisdom of DHGC, CSA first used DHGC to adaptively model the boundary information in low-level features. Then, we use the LCA module (our simplified version of CA) to establish long-range dependencies by encoding the features along with horizontal and vertical coordinates at each pixel. The encoded low-level features are introduced into the high-level features of the decoder network to refine the boundaries of remote sensing images and improve segmentation performance.

Based on the above considerations, we propose a novel boundary-sensitive network (BSNet), which is based on the popular encoder–decoder architecture [11], [12]. BSNet can effectively improve the segmentation quality of remote sensing images. The primary contributions of this work are given as follows.

1) We propose GCKs to explicitly encode boundary information. GCKs jointly model traditional boundary detection operators and CNN-based methods. Furthermore, we boost GCKs into DHGC to replace VC, which aggregates one VCK and two GCKs through an adaptive strategy to ensure that boundary information and semantic information can be fully extracted simultaneously in the encoder network.
2) We apply the CSA in the decoder network. The CSA module sequentially uses the DHGC and the LCA module to adaptively model boundary information and long-range dependencies in low-level features, helping high-level features to recover clearer and sharper boundaries.
3) We have developed the BSNet framework to extract and recover complete and clear boundaries of remote sensing images. Experiments on three classical remote sensing datasets, the ISPRS Vaihingen, Potsdam benchmarks, and the iSAID dataset, demonstrate that BSNet achieves better performance than prevalent methods.

The remainder of this article is organized as follows. Section II introduces the related work. Section III describes the proposed BSNet architecture and details of the components. The experimental setup and results, as well as the ablation studies, are presented in Section IV. Section V summarizes this work.

II. Related Works

A. Remote Sensing Image Semantic Segmentation

Semantic segmentation has attracted much attention as a fundamental issue due to its significance in a variety of applications in remote sensing. The traditional methods were based on region merging, such as SLIC [42] and K-means [43], where handcrafted features were extracted and then merged. Over the past years, inspired by the successful development of deep learning in computer vision, convolutional neural networks (CNNs) have been applied to remote sensing image semantic segmentation with powerful segmentation ability.

The FCN [11] is the pioneering and successful attempt to use CNN for semantic segmentation. Despite the promising performance of the FCN and its variants [44]–[46] in semantic segmentation tasks, the downsampling operations adopted by them lose details, resulting in blurred boundaries in the segment and the inability to detect small objects. The loss of detail and spatial resolution induced by downsampling can be alleviated by skip connections [47], [48] and contextual feature extraction modules [24], [49]. In [25], [26], and [49]–[51], dilated convolution [52] was employed to aggregate more contextual features. However, such a sparse convolution may induce grid distortions when used successively [53]. Later, skip connections [47], [48] were proposed to combine low-level features into the decoder network, showing their advantage in remote sensing image semantic segmentation [54].

Furthermore, the scale of objects in remote sensing images is inconsistent, unlike in the natural images taken at a close range. Accordingly, several works were conducted to improve the semantic segmentation for remote sensing images, considering the scale inconsistency property. In [55]–[59], more contextual information was extracted from remote sensing images. In addition, other deep learning methods are studied to improve the quality of the semantic segmentation boundaries of remote sensing images. ERN [15] adopted the spatial boundary context to alleviate semantic ambiguity. The SBANet [14] was proposed to effectively generate and apply refined land-cover boundaries.

B. Boundary Detection

Boundary detection plays a key role in various computer vision problems [60]. In the conventional methods, boundary detection operators, such as Sobel [61] and LoG, were used to detect low-level boundaries. HED [62] learned hierarchical boundary representation for learning guides. CASENet [60] formulated boundary detection as a multilabel problem, where each boundary pixel can be labeled with more than one label. Motivated by the advantage of boundary information in refining segments, edge information has been explored in several studies to enhance semantic segmentation boundaries. The CNN-learned edge was used to improve the semantic segmentation in [63]. In [55], the first study was conducted to adopt edge constraints in learning a deep semantic segmentation network for satellite imagery. Liu *et al.* [15] proposed a novel edge loss reinforced semantic segmentation network (ERN) to eliminate semantic ambiguity via spatial boundary context. The ERN simultaneously achieves boundary detection and semantic segmentation without a significant model complexity increase. A superpixel algorithm was used to extract rich boundary information in [64]. EANet [28] designed an edge-aware multitask learning model and EALoss to extract buildings. FusionNet [65] designed an edge-aware DCNN to address semantic segmentation of land, sea, and ship. Bertasius *et al.* [66] introduced a boundary neural field (BNF), an architecture that employs a semantic segmenta-

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tion FCN to predict semantic boundaries and then uses the predicted boundaries and the FCN output to produce an improved semantic segmentation maps a global optimization. Chong *et al.* [67] consider edge and context as the key to solving the problem of semantic segmentation of small-scale objects in VHR remote sensing images. Thus, they divide the network into semantic streams and edge streams. Li *et al.* [68] propose a novel framework to improve the semantic segmentation results by decoupling features into the body and the edge parts to handle inner object consistency and fine-grained boundaries jointly. Sun *et al.* [29] proposed BAS^{4}Net for the semantic segmentation of VHR remote sensing images. It can learn additional information related to object boundaries in a semisupervised manner. Takikawa *et al.* [20] designed a gated convolution layer (GCL) to focus on boundary information and add a shaped stream using GCL to capture the feature maps particularly relevant to the boundaries.

C. Dynamic Convolution

In the conventional convolutional layers [69], [70], the number of convolution kernels is fixed. However, the fixed kernel of convolution could be inefficient on the hardware, particularly in the image domain. In natural language processing (NLP) fields, several works [71]–[73] were conducted to incorporate context to establish input-aware convolution filters. It was varied according to input sentences with various lengths. Inspired by the promising adaptivity and flexibility in NLP tasks, dynamic convolution was proposed in [74]–[78]. The dynamic convolution predicts the coefficients used to combine fixed kernels, achieving lighter computation on hardware. The dynamic convolution increases the model capability by aggregating multiple convolution kernels via attention while retaining the depth and width of the network. The dynamic convolution kernels are assembled differently for different input images.

D. Attention Module

The attention modules have been recently widely used in various tasks due to their promising ability to enhance performance with only a few additional parameters [79]. The channel attention module was proposed in SENet [80] to weigh beneficial channels while suppressing less important channels. CBAM [81] extended this idea to spatial domains with spatial information encoding via large kernel convolutions. The idea was also extended by adopting different advanced attention blocks in GENet [82], GALA [83], AA [84], and CA [41]. The spatial pyramid was combined into the attention mechanism instead of atrous convolutions in PAN [85]. In recent years, nonlocal attention networks have been actively studied because of their ability to construct spatial attention and channel attention. GCNet [86], A2Net [87], NLNet [88], SCNet [89], GSoP-Net [90], and CCNet [91] proposed different types of nonlocal mechanisms to capture different spatial information. Attention modules have been used to extract enhanced features for semantic segmentation. For instance, the network attended the origin feature with the attention modules that take local details and boundary information as input.

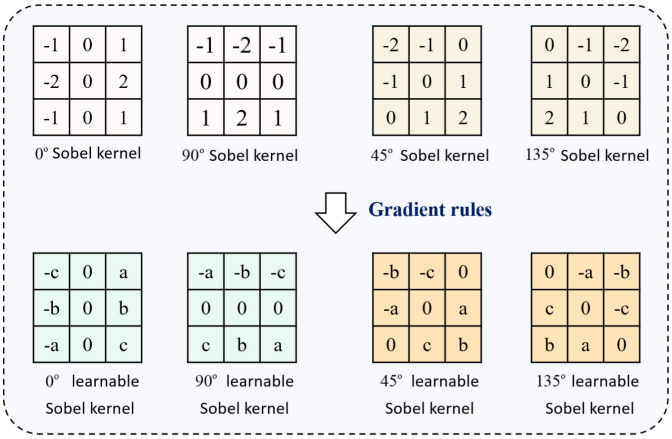


Fig. 3. Learnable Sobel kernels. The learnable kernel corresponding to each Sobel kernel is derived according to the gradient rule.

III. Proposed Method

To extract and recover boundary information effectively, three novel modules are proposed: GCKs, DHGC, and CSA. GCKs and DHGC are proposed to extract rich boundary information in the encoder network, and CSA is developed to recover fine boundary information in the decoder network. In this section, we first introduce these three modules and then fuse them to build the proposed BSNet.

A. Gradient Convolution Kernels

The boundary information extracted by VC gradually vanishes as continuous downsampling. This leads to significant errors in boundary prediction. As mentioned above, our goal is to enable the convolution kernels to enhance the extraction of boundary information. Therefore, we customized the GCKs to extract the boundary information with sharp gradient changes. The VC is formulated as follows.

For the VCK, 2-D convolution includes two steps: 1) sampling with grid R on the input feature maps $\mathbf{x}$ and 2) weighted summation of sampled values with $\mathbf{w}$. The receptive field and dilation ratio are determined by the grid R. For example,

$$R = \{(-1,-1),(0,-1),\ldots,(0,1),(1,1)\} \tag{1}$$

define a 3×3 kernel with one dilation.

For each position $\mathbf{p}_0$ on the output feature map $\mathbf{y}$

$$\mathbf{y}(\mathbf{p_0}) = \sum_{\mathbf{p_n} \in R} \mathbf{w}(\mathbf{p_n})\mathbf{x}(\mathbf{p_0} + \mathbf{p_n}) \tag{2}$$

where p_n enumerates the positions in R.

The GCKs are inspired by the traditional Sobel operator. The Sobel operator [61] has been applied as *a priori* template in semantic segmentation based on deep learning. This operator consists of two pairs of the most commonly used 3×3 kernels (0°–90° Sobel kernels and 45°–135° Sobel kernels), as shown in Fig. 3. These kernels are designed to generate maximum responses at gradient directions of 0°, 90°, 45°, and 135°. For example, 0°–90° Sobel kernels in grid R can be expressed as

$$\begin{aligned} S_{0^\circ} &= \{-1,0,1,-2,0,2,-1,0,1\} \\ S_{90^\circ} &= \{-1,-2,-1,0,0,0,1,2,1\}. \end{aligned} \tag{3}$$

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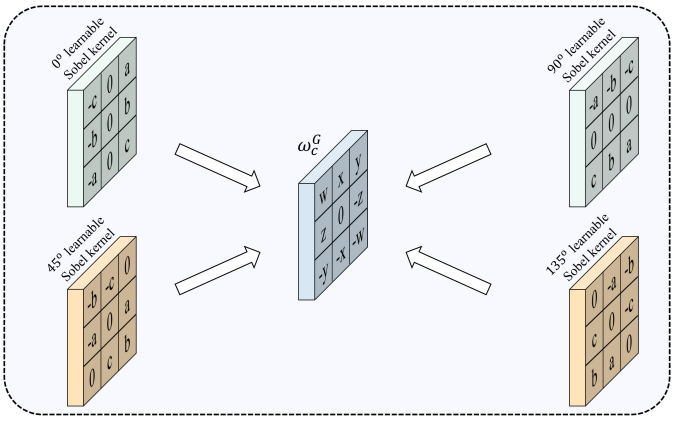


Fig. 4. GCK ($\mathbf{w^G}$) on any cth channel. It integrates the gradient rules of the 0°, 90°, 45°, and 135° learnable Sobel kernels.

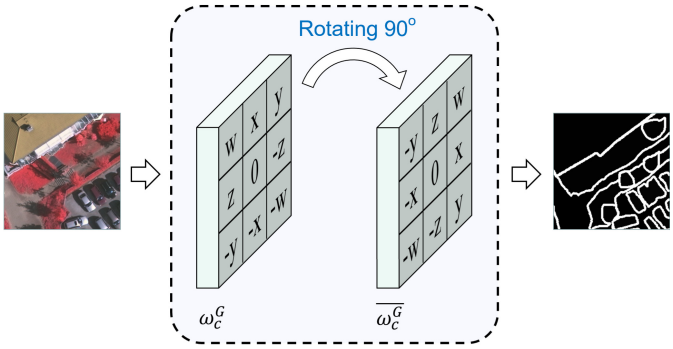


Fig. 5. GCKs. GCKs include $\mathbf{w^G}$ and $\overline{\mathbf{w^G}}$, and $\overline{\mathbf{w^G}}$ is obtained by rotating $\mathbf{w^G}$ by 90°. GCKs are committed to extracting boundary information with obvious gradient changes.

However, the Sobel operator is unlearnable, and its fitting ability is not powerful for semantic segmentation of remote sensing images with a large amount of data. According to the gradient rule of Sobel kernels, the learnable kernels are designed, as shown in Fig. 3. The learnable Sobel kernels corresponding to 0°–90° can be described as

$$\begin{aligned} \mathrm{LS}_{0^\circ} &= \{-c, 0, a, -b, 0, b, -a, 0, c\} \\ \mathrm{LS}_{90^\circ} &= \{-a, -b, -c, 0, 0, 0, c, b, a\}. \end{aligned} \tag{4}$$

Unlike the traditional Sobel operator, the learnable Sobel operator does not require special definitions for the values of a, b, and c. Any one of a, b, and c can be positive or negative, and there is no need to guarantee the maximum value of b. The learnable Sobel operator satisfies that, when the convolution kernel acts on the interior of the target, its response is 0, and when it acts on the boundaries, its response is not 0, thereby capturing the boundary information with sharp gradient changes.

The learnable kernels sensitive to more gradient directions can help extract rich boundary features. Therefore, we are committed to jointly modeling the Sobel operator and deep learning methods, and propose learnable gradient convolution that is sensitive to more gradient directions.

As shown in Fig. 4, one of the GCKs ($\mathbf{w^G}$) integrates the gradient rules of the 0°–90° and 45°–135° learnable kernels. In other words, each of these kernels is a subset of the $\mathbf{w^G}$. Therefore, the kernel has a more sensitive ability to capture the boundaries of a large number of input images in multiple directions. As shown in Fig. 5, the other kernels are obtained by rotating 90°, and their sensitive gradient directions are vertical.

GCKs are based on one VCK; the other kernels $\mathbf{w^G}$ and $\overline{\mathbf{w^G}}$ in a regular grid are constrained to be on any cth channel to satisfy

$$\begin{aligned} \omega_c^G &= \{w, x, y, z, 0, -z, -y, -x, -w\} \\ \overline{\omega_c^G} &= \{-y, z, w, -x, 0, x, -w, -z, y\} \end{aligned} \tag{5}$$

which defines a set of 3×3 kernels. $\overline{\mathbf{w^G}}$ is obtained by rotating $\mathbf{w^G}$ by 90°.

In short, compared with VC, GCKs can effectively capture boundary information with sharp gradient changes.

B. Dynamical Hybrid Gradient Convolution

Although the GCKs can capture boundary information with sharp gradient changes, they cannot capture rich semantic information. Therefore, how to adaptively combine GCKs for boundary information and VCKs for semantic information is an urgent problem to be solved. Therefore, we propose dynamical hybrid gradient convolution (DHGC). The general framework of the DHGC is illustrated in Fig. 6.

DHGC aggregates the HGCKs and weight coefficients through the kernel aggregation module. The HGCKs are composed of one VCK and two GCKs. The GCKs are customized to extract the boundary information with sharp gradient changes. The VCK ensures the extraction of semantic information. Parallel HGCKs can improve the extraction of boundary information while ensuring the extraction of semantic information. Since the boundary information in different input images is variable, the CPM generates adaptive coefficients for one VCK and two GCKs according to the image content to improve the expression ability of the network. Thus, DHGC is an adaptive kernel sensitive to boundary information. This part first introduces the CPM in detail and then aggregates CPM and HGCKs into a high-efficiency operator through the kernel aggregation method.

1) Coefficient Prediction Module: Inspired by the attention module [75], [80], the CPM predicts the weight coefficients (η_i) of the VCK and GCKs according to the content of each input image ($\mathbf{I}$), thereby adaptively enhancing the feature capture capability.

As shown in Fig. 7, global average pooling (GAP) is first adopt to aggregate the global spatial information of the entire input feature map ($\mathbf{I}$) into $1 \times 1 \times C_{\text{in}}$ vector. Then, we use the full connection (FC) layer to map the extracted vectors to $1 \times 1 \times C$ ($C = 3$, since HGCKs are composed of three kernels) vectors. In the end, the vector $1 \times 1 \times C$ is processed by the softmax function (Softmax) to obtain the weight coefficients ($\boldsymbol{\eta}$). The Softmax constrains the sum of the coefficients to 1, which greatly simplifies the learning of coefficients. In other words, (η_i) controls the contribution of GCKs. When (η_1)

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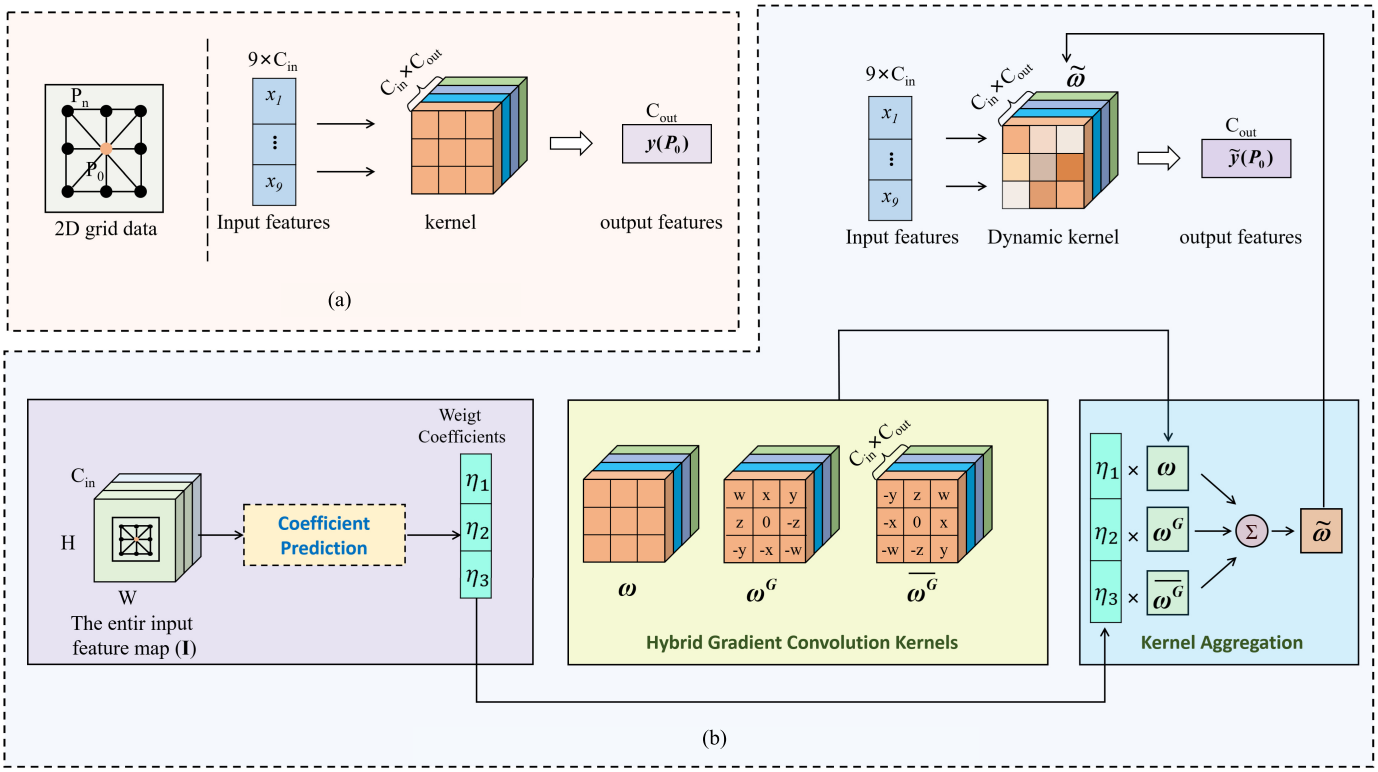


Fig. 6. Overall framework of DHGC. It aggregates HGCKs dynamically based upon input-dependent coefficients. Sections III-A and III-B describe the method in more detail. (a) 2-D 3 × 3 convolution. (b) 2-D 3 × 3 dynamic hybrid convolution.

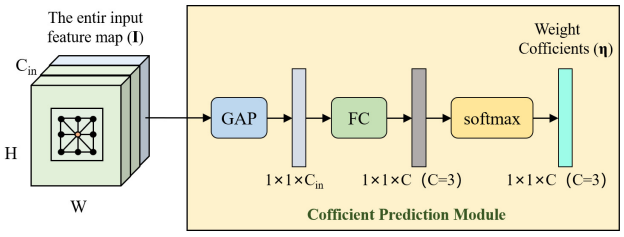


Fig. 7. CPM.

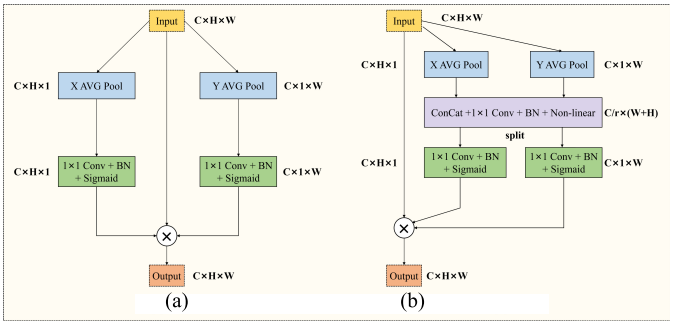


Fig. 8. Details of (a) LCA and (b) CA.

and (η_2) are equal to 0 in the extreme case, DHGC degenerates into a VCK. These processes can be formulated as follows:

$$\boldsymbol{\eta} = \text{Softmax}(\text{FC}(\text{GAP}(\mathbf{I}))). \tag{6}$$

In order to extract the boundary information more effectively, we adopt the method of kernel aggregation. The specific details are described in the following.

2) Kernel Aggregation: The kernel aggregation module aggregates the weight coefficients and HGCKs to generate DHGC. We denote the HGCKs as $\{\mathbf{w}, \mathbf{w^G}, \overline{\mathbf{w^G}}\}$, the DHGC as $\widetilde{\mathbf{w}}$, and the coefficient as η_k, where $k = 0, 1, 2$.

Once the coefficients are obtained, $\widetilde{\mathbf{w}}$ can be generated by the following formula:

$$\widetilde{\mathbf{w}} = \eta_0 \mathbf{w} + \eta_1 \mathbf{w^G} + \eta_2 \overline{\mathbf{w^G}}. \tag{7}$$

Kernel aggregation is only used to generate DHGC during inference. It is not suitable for batch-based training methods. Because the weights of DHGC in the same minibatch are not shared, different inputs correspond to different weights. Therefore, we fused the output feature maps and the weight coefficients instead of performing kernel aggregation during the training time. They are mathematically equivalent, as shown in the following equation:

$$\begin{aligned} \widetilde{\mathbf{w}} \circledast \mathbf{x} &= (\eta_0 \mathbf{w} + \eta_1 \mathbf{w^G} + \eta_2 \overline{\mathbf{w^G}}) \circledast \mathbf{x} \\ &= (\eta_0 \mathbf{w}) \circledast \mathbf{x} + (\eta_1 \mathbf{w^G}) \circledast \mathbf{x} + (\eta_2 \overline{\mathbf{w^G}}) \circledast \mathbf{x} \\ &= \eta_0 \mathbf{y} + \eta_1 \mathbf{y^G} + \eta_2 \overline{\mathbf{y^G}} \end{aligned} \tag{8}$$

where $\circledast$ represents the convolution operation, and $\mathbf{y^G}$ and $\overline{\mathbf{y^G}}$ represent the output feature maps of $\mathbf{w^G}$ and $\overline{\mathbf{w^G}}$, respectively. Similar to (2), $\mathbf{y^G}$ and $\overline{\mathbf{y^G}}$ can be calculated as follows:

$$\begin{aligned} \mathbf{y^G}(\mathbf{p_0}) &= \sum_{\mathbf{p_n} \in \mathbf{R}} \mathbf{w^G}(\mathbf{p_n}) \mathbf{x}(\mathbf{p_0} + \mathbf{p_n}) \\ \overline{\mathbf{y^G}(\mathbf{p_0})} &= \sum_{\mathbf{p_n} \mathbf{R}} \overline{\mathbf{w^G}(\mathbf{p_n})} \mathbf{x}(\mathbf{p_0} + \mathbf{p_n}). \end{aligned} \tag{9}$$

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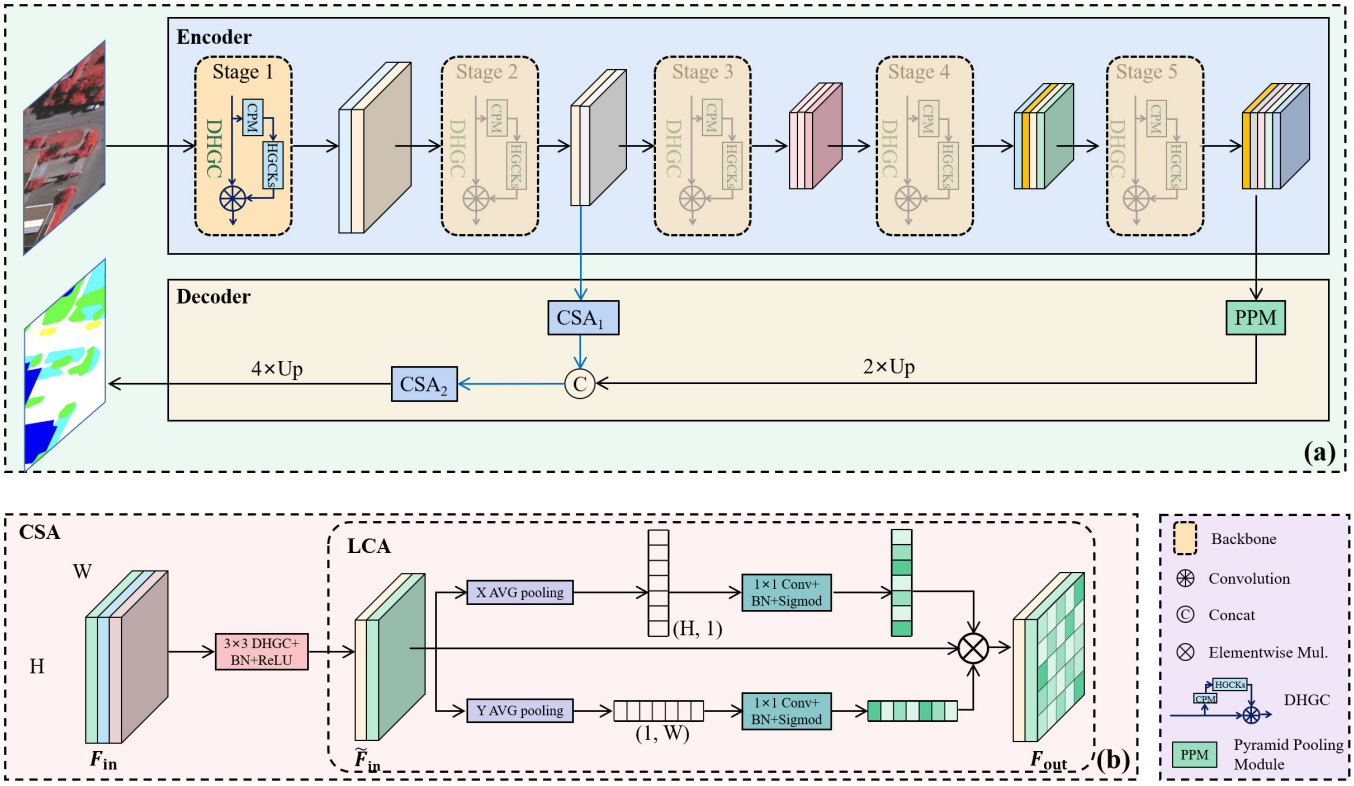


Fig. 9. Structure of the proposed BSNet and the CSA. (a) Encoder network of BSNet applies DHGC to enhance the extraction of boundary information. Subsequently, the boundary-enhanced feature maps are fed into the PPM to extract contextual information. Then, the CSA is embedded in the network to refine boundary predictions. (b) CSA adaptively models the boundary information in the low-level features and further uses the LCA module to mine its long-range dependencies, helping to enrich the boundary details of high-level features in the decoder network and recovering finer segmentation results.

Kernel aggregation allows DHGC to simplify mathematically more expensive expressions during inference. Therefore, DHGC is a boundary information enhancement method with high computational efficiency.

C. Coordinate Sensitive Attention Module

In the stage of features recovery, the high-level features of the decoder network lack spatial detail information, which can also cause blurred boundary prediction. Skip connections are unreasonable to directly introduce low-level features into high-level features to recover detailed information because they do not model the long-range dependencies of low-level features that are critical to recovering fine boundaries. In addition, the complexity of the boundaries of different input images is different. How to adaptively model the boundary information in the low-level features and introduce it into the high-level features is also an urgent issue to be solved. We propose CSA to solve these issues. The modeling process is mainly divided into two steps. The first step is to use DHGC adaptively to enhance the extraction of boundary information, and the second step is to use relatively lightweight LCA (the details are shown in Fig. 8) to model the long-range dependencies of feature maps after the enhancement of boundary information. The LCA is our simplified version, which is based on the recently popular strip pooling [92] and CA [41]. Different from CA, LCA removes the redundant concat layer and a 1×1 convolution layer in CA. Next, we first outline the characteristics of LCA.

The LCA module performs long-range pooling along the X-direction and the Y-direction on the feature maps and then multiplies the obtained results with the original feature maps according to the pixel position to make each pixel in the feature maps obtain long-range dependencies. Suppose that the result of long-range pooling in the X-direction is expressed as $\mathbf{g^h}$ after 1×1 convolutional encoding, and the same in the Y-direction is expressed as $\mathbf{g^w}$. For the input $\widetilde{\mathbf{F}}_{\mathbf{in}}$ of LCA, the output $\mathbf{F_{out}}$ of LCA can be formulated as

$$f_{\text{out}c} = \widetilde{f}_{\text{in}c}(i, j) \times g_c^h(i) \times g_c^w(j). \tag{10}$$

$\mathbf{g^h}$ and $\mathbf{g^w}$ embed the captured long-range dependencies in the X-direction and the Y-direction into the feature maps by multiplying with the original feature maps.

Our proposed CSA is shown in Fig. 9(b). CSA has three main advantages over CA. First, CSA uses DHGC to adaptively enhance the boundary information in feature maps and then establishes the long-range dependencies for the feature maps with enhanced boundary information. In contrast, CA only establishes the long-range dependencies for the common feature maps without optimizing for the boundary. Second, the CSA we studied considered the lightness of the method. We use DHGC to not only enhance the adaptive boundary information for the feature maps but also control the

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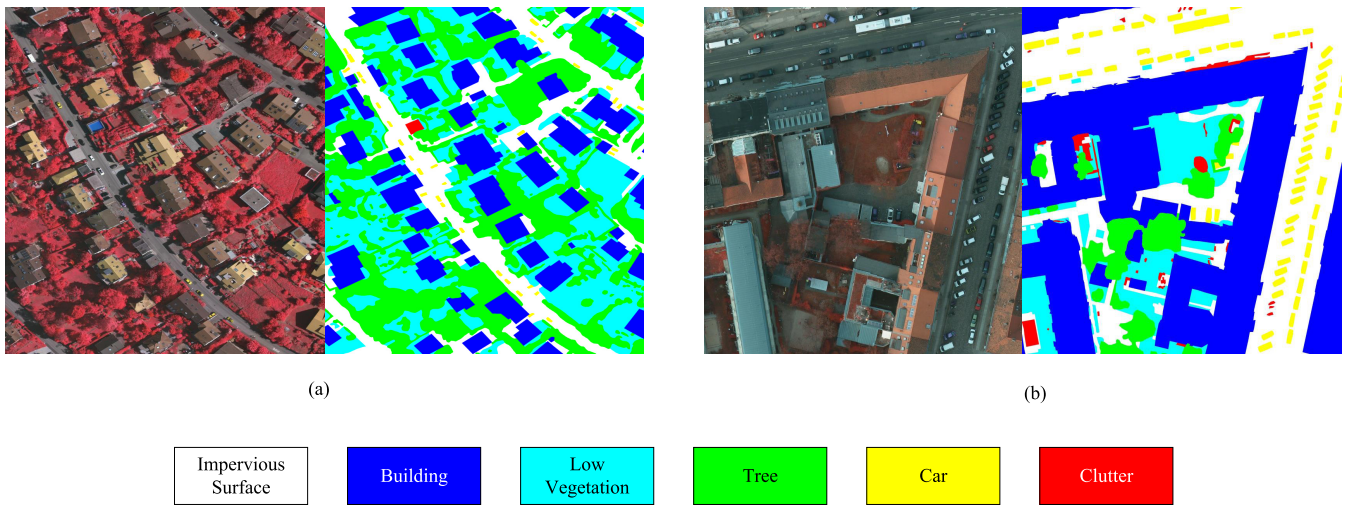


Fig. 10. Image sample and corresponding ground truth of (a) Vaihingen and (b) Potsdam challenges. The labels of Vaihingen and Potsdam challenges include six categories: impervious surface (white), building (blue), low vegetation (cyan), tree (green), car (yellow), and background (red).

channel dimensions from $\mathbf{F_{in}}$ to $\widetilde{\mathbf{F}}_{\mathbf{in}}$ to reduce the redundancy of the network. This method was inspired by [93]. Third, it should be emphasized that the LCA module is a simplified version of CA. This improvement enhances the timeliness of the network while ensuring the performance of the model. Subsequently, we will use the X-direction and the Y-direction long-range pooling for the feature $\widetilde{\mathbf{F}}_{\mathbf{in}}$ modeled by DHGC to assist in establishing the long-range dependence between pixels in the feature. In conclusion, CSA can be used to adaptively enhance boundary details and establish long-range dependencies between pixels.

D. BSNet for Segmentation

The general structure of BSNet is displayed in Fig. 9(a). Considering the effectiveness of PSPNet [24], we use it as the baseline network for semantic segmentation tasks. BSNet is mainly made out of three structures: the encoder network, the PPM, and the decoder network.

In the encoder network, the remote sensing images are input into the backbone to extract the semantic features. The 3×3 convolution kernels in the backbone network (ResNet series) are replaced by DHGC to generate boundary-information enhanced feature maps. We follow the previous method of PSPNet that consisted of five stages (Stages 1–5) to extract intensive information in the backbone. Then, the semantic features are fed into the PPM module [24] due to its excellent ability to capture contextual information.

In the decoder network, we only introduce the low-level features of Stage 2 to assist high-level features to recover detailed information. This is because feature maps of Stage 3 and 4 have the same resolution as feature maps of Stage 5. Spatial detail information is closely related to the resolution. Therefore, considering that, if the feature maps of each stage in the encoder network are introduced into the corresponding decoder network, it will bring a lot of feature redundancy, inspired by Deeplab v3+, we only introduce the feature maps of Stage 2 and then concatenate them with the features upsampled from the PPM module. This is due to that the feature maps of Stage 2 have a larger resolution and, thus, have more detailed information. Different from Deeplab V3+, we use CSA at two nodes. As shown in Fig. 9(a), the CSA_1 predicts the boundaries of category pixels more accurately by adaptively modeling the boundary information and long-range dependencies in the low-level features from Stage 2 and then introducing them into the high-level features. After the encoded low-level features are connected with the upsampled high-level features, CSA_2 further adaptively models its boundary details. Subsequently, the outputs of CSA_2 are processed through 1×1 convolutional layers and upsampling operators. Generate output feature maps whose resolution and output channels are consistent with the input image and the number of categories, respectively.

BSNet promotes the awareness of complex boundaries of remote sensing images and generates fine segmentation results. All of the parameters in the BSNet can enable adaptive end-to-end learning from the data.

IV. Experimental Results

A. Experimental Setting

1) Datasets: The effectiveness of the proposed BSNet is evaluated on three datasets: ISPRS Vaihingen, Potsdam benchmarks, and the iSAID dataset. Some examples of images are shown in Fig. 10.

The Vaihingen dataset includes 33 images for which the spatial resolution is 9 cm, and the average image size is 2494×2064. Each image is composed of three channels: red, green, near-infrared (NIR), and the corresponding digital surface model. The Vaihingen dataset has six classes (impervious surfaces, buildings, low vegetation, trees, cars, and background). Networks are trained and validated using 16 images with ground truth, while the remaining 17 images are used for testing. Note that the digital surface model is not used in our experiments.

The Potsdam dataset includes 38 images for which the spatial resolution is 5 cm, and the image size is 6000×6000. Each image is composed of four channels: red, green, blue, NIR,

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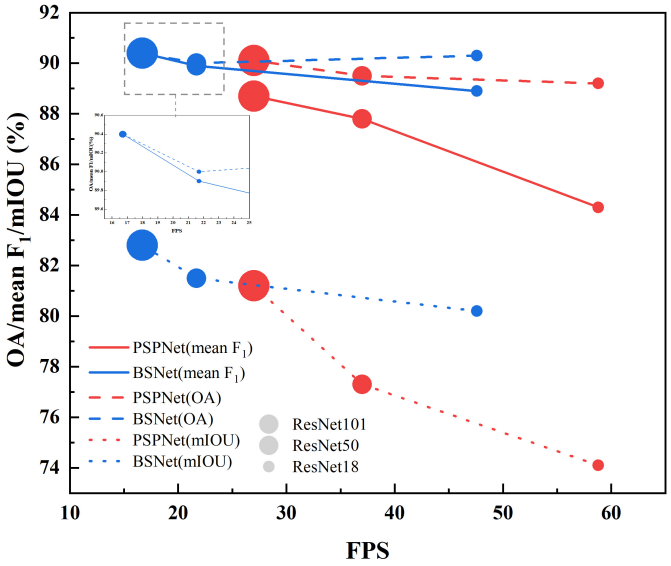


Fig. 11. Accuracy and inference speed on different backbones of BSNet and PSPNet (baseline).

the corresponding digital surface model, and the normalized digital surface model. All images are densely labeled in the same six classes as the Vaihingen dataset. Networks are trained and validated using 24 images with ground truth, while the remaining 14 images are used for testing. In order to have a fair comparison with the previous work [58], we followed its experimental setup; note that only NIR, red, and green channels are used in our experiments.

The iSAID dataset consists of 2806 high-resolution airborne images with a spatial dimension from 800 × 800 pixels to 4000 × 13 000 pixels. The iSAID dataset consists of 655 451 annotated instances of 15 foreground categories. Similar to [94], 1411 images are used for training, and 458 images are used for evaluation. Note that the remaining 937 images in the test are unavailable.

2) Network Training: PSPNet [24] is used as the baseline. And ResNet is employed as the backbone. All networks are implemented with PyTorch.

Vaihingenand and Potsdam Datasets: The training and testing samples are cropped with a 256 × 256 sliding window from the training and testing images. A stride of 64 is used for sampling the training samples, while no stride is used for sampling the testing samples. The training samples are augmented with random scaling and flipping. The networks are trained with a stochastic gradient descent (SGD) optimizer and batch size = 6. An initial learning rate of 10^{-2} with a polynomial decay policy is used.

iSAID Dataset: The training and validation samples are cropped with an 896 × 896 sliding window from the training and testing images. A stride of 512 is used for sampling the training samples, while no stride is used for sampling the testing samples. The training samples are augmented with random scaling and flipping. The networks are trained with an SGD optimizer and batch size = 8. An initial learning rate of 0.007 with a polynomial decay policy is used.

All the momentum is set to 0.9, and the initial weight decay is set to 0.0001. All backbones are pretrained on the ImageNet dataset [95] due to the performance contribution of an initialization [96].

3) Evaluation Metrics: To evaluate the performance of the proposed BSNet, four common metrics are used: the overall accuracy (OA), the F_1 score, the mean of classwise intersection over union (mIoU), and the accuracy of foreground categories (Acc.). The OA is used to assess the proportion of correctly classified pixels to the total pixels. For each category, the F_1 score can be computed by

$$F_1 = (1+\beta^2)\cdot\frac{\text{precision}\cdot\text{recall}}{\beta^2\cdot\text{precision}+\text{recall}},\quad \beta=1. \tag{11}$$

Furthermore, all the F_1 scores are averaged to compute the mean F_1 score. The accuracy of foreground categories (Acc.) is used for the iSAID dataset with extremely imbalanced foreground and background.

4) Backbone Setting: Fig. 11 records the accuracy and inference speed of BSNet and PSPNet (baseline) on different backbones. Compared with the baseline under different backbones, BSNet shows stable and significant advantages with an ignorable extra computational cost. Although the increase in the network depth can improve prediction accuracy, its efficiency has dropped sharply. In particular, the accuracy gain of BSNet compared to PSPNet decreases as the depth of the backbone increases. Note that the accuracy of BSNet (ResNet-18) is better than that of PSPNet (ResNet-101), and the inference speed is approximately 1.7 times higher. Therefore, considering the balance of accuracy and efficiency and to speed up experiments, we employ ResNet-18 as our backbone in the ablation studies.

B. Ablation Studies

Ablation studies are performed on the Vaihingen dataset to verify the superiority of our method. In the following ablation studies (1) and (2), in order to verify the effectiveness of the GCKs and CPM, we only embed DHGC or its counterpart into BSNet.

1) Effect of Gradient Convolution: To evaluate the performance improvement of GCKs, we compared DHGC and its counterparts, as shown in Fig. 12. We replace all of the VCKs of the baseline with DHGC and its counterparts, and then verify the performance of these networks. We, finally, present the accuracy in Table I. Through this modification, the mean F_1 score of the baseline network has been improved to varying degrees, and DHGC shows the best performance. The improvement of the boundary F_1 score is more obvious, which proves the effectiveness and superiority of our proposed GCKs.

Furthermore, as shown in Fig. 13, two pairs of traditional Soble kernels and three kinds of learnable Sobel kernel orders have been experimentally verified: 1) 0°–90° Sobel kernels; 2) 45°–135° Sobel kernels; 3) 0°–90° learnable order, following the definition of the pair of 0°–90° Sobel kernels; 4) 45°–135° learnable order, following the definition of the pair of 45°–135° Sobel kernels; and 5) GCKs order, meaning the approximation of the Sobel operators (our work). We verify

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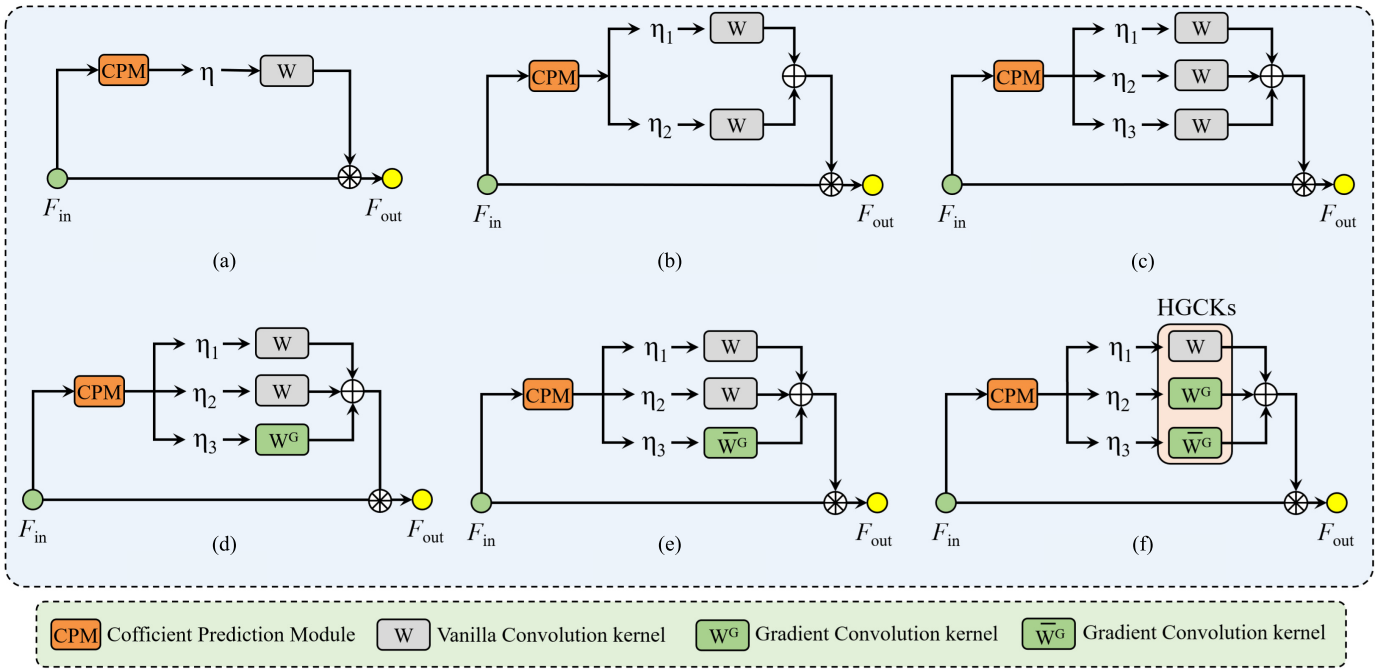


Fig. 12. Counterparts to compare against DHGC. (a) Single kernel. (b) Double kernels. (c) Triple kernels. (d) $HGCK_1$. (e) $HGCK_2$. (f) DHGC.

TABLE I
COMPARATIVE EXPERIMENT RESULTS OF DHGC AND ITS COUNTERPARTS

Method	Kernel	mean $F_1(\%)$	5px $F_1(\%)$	7px $F_1(\%)$
PSPNet (baseline)	–	84.3	53.8	56.9
BSNet	Single Kernel	84.6	54.0	57.1
	Double Kernels	85.1	55.2	58.2
	Triple Kernels	85.6	55.8	58.8
	$HGCK_1$	86.7	57.1	59.6
	$HGCK_2$	86.6	57.3	60.1
	DHGC	**87.6**	**58.0**	**60.6**

TABLE II
PERFORMANCE COMPARISON OF SOBEL KERNELS AND DIFFERENT LEARNABLE SOBEL KERNEL ORDERS

Method	mean $F_1(\%)$	5px $F_1(\%)$	7px $F_1(\%)$
$0^\circ - 90^\circ$ Sobel kernels	85.3	55.1	58.4
$45^\circ - 135^\circ$ Sobel kernels	85.5	55.6	58.2
$0^\circ - 90^\circ$ learnable order	86.7	56.4	58.7
$45^\circ - 135^\circ$ learnable order	86.5	56.1	58.6
GCKs learnable order	**87.6**	**58.0**	**60.6**

the superiority of GCKs by replacing the GCKs of DHGC with the remaining four operators. The results are shown in Table II.

On the one hand, in terms of the traditional Sobel kernels compared with the learnable Sobel orders, experimental results show that the traditional Sobel kernels have a certain performance improvement compared with the baseline, but their performance improvement is far less than that of the learnable Sobel orders and not as strong as our GCks. This is because traditional Sobel kernels use fixed local filters to aggregate gradient information every time, and these local filters may not be optimal and suitable for the current task. However, the learnable Sobel order can be data-driven to refine the weights

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TABLE III
COMPARISON VERSUS BASELINE AT DIFFERENT THRESHOLDS IN TERMS OF BOUNDARY F_1-SCORE ON THE VAIHINGEN DATASET

Thresh.	Method	Impervious Surface	Building	Low Vegetation	Tree	Car	mean F_1(%)
5px	PSPNet	58.5	62.2	44.9	53.4	49.9	53.8
	BSNet	**62.2**	**64.6**	**51.0**	**56.1**	**59.0**	**58.0**
7px	PSPNet	61.9	64.6	48.3	55.1	54.6	56.9
	BSNet	**64.1**	**66.8**	**52.6**	**57.6**	**61.8**	**60.6**
9px	PSPNet	63.1	67.4	50.5	57.2	59.3	59.5
	BSNet	**66.3**	**69.2**	**54.4**	**59.4**	**65.2**	**62.9**
12px	PSPNet	65.4	69.5	52.5	58.8	62.8	61.8
	BSNet	**68.3**	**71.3**	**56.2**	**61.2**	**68.1**	**65.0**

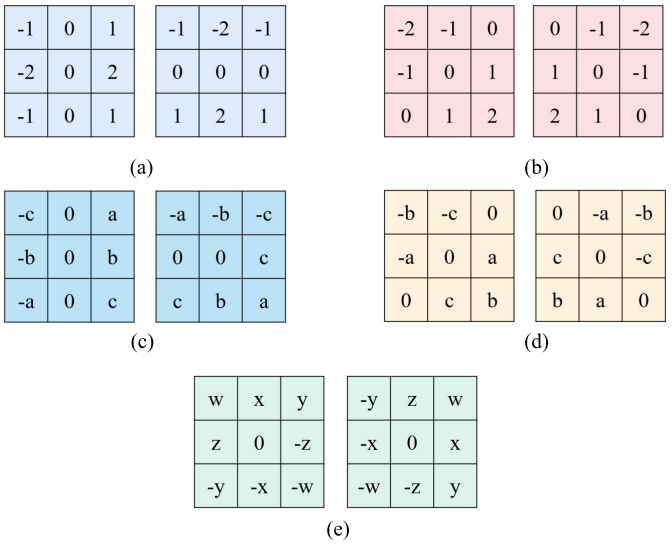


Fig. 13. Overview of two pairs of traditional Sobel kernels and three kinds of learnable Sobel orders. (a) 0°–90° traditional Sobel kernels. (b) 45°–135° traditional Sobel kernels. (c) 0°–90° learnable Sobel order. (d) 45°–135° learnable Sobel order. (e) GCKs learnable order.

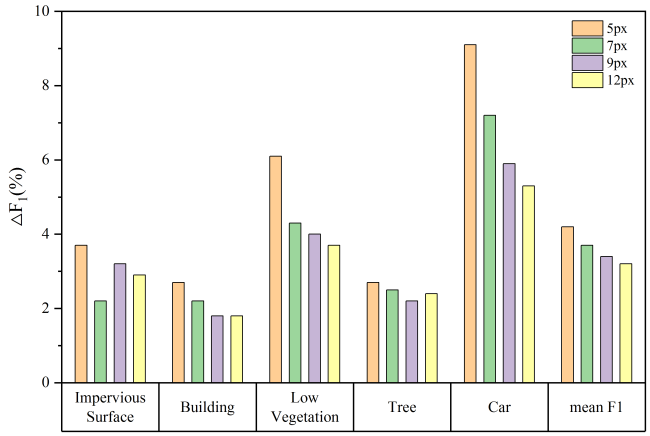


Fig. 14. Comparison of performance improvement at different thresholds.

of the convolution kernel, which makes the learnable Sobel more powerful in data fitting and presentation. On the other hand, in terms of learnable Sobel orders versus our GCKs, if the definition of the Sobel kernels is strictly followed, then the setting of the learnable Sobel kernels order should be of the 0°–90° learnable order or 45°–135° learnable order, but we found that the performance of our GCKs is better than the above two settings. The reason is that our GCKs are more sensitive to the extraction of multiple directions from a large number of input feature maps.

To quantitatively verify the validity of the proposed method, we directly compare the performance of BSNet and baseline at boundary pixels. With the method mentioned by Takikawa *et al.* [20], we have compared the performance with boundary thresholds of 5, 7, 9, and 12 px, respectively. Table III shows that BSNet has a significant accuracy improvement in terms of boundary accuracy compared with the baseline. Furthermore, to verify that our method improves performance at boundary pixels more than object internal pixels, we compared the performance improvements at 5, 7, 9, and 12 px. Fig. 14 illustrates that the performance advantage increases as the boundaries become narrower, which directly proved that our method was more efficient in boundary pixels. Comparing the performance improvements of different categories, Car with a larger proportion of boundary pixels and Low Vegetation with indistinguishable boundaries have more significant improvements. The results prove that BSNet is, indeed, effective at the boundaries.

The fourth ablation experiment that we performed on GCKs is the visualization of the output features of the GCKs. Fig. 15 shows some visualization examples. For the GCKs, the object boundaries are highlighted, which means that the GCKs pay more attention to boundary pixels. This confirmed that the GCKs did, indeed, capture the expected boundary information. In contrast, the VCK pays more attention to global information. It shows that DHGC can flexibly store boundary information and global information, which can produce clearer

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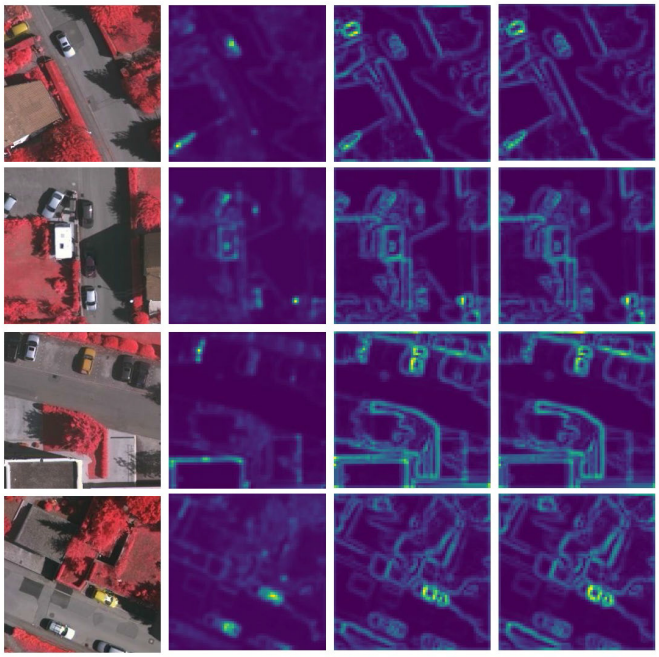


Fig. 15. Visualization of output feature maps from different kernels in Stage 2. Columns show the input images, the output feature maps of VCK ($\mathbf{w}$), GCK ($\mathbf{w}^{\mathbf{G}}$), and GCK ($\overline{\mathbf{w}^{\mathbf{G}}}$).

TABLE IV
ABLATION EXPERIMENT RESULTS ON CPM AND GCKS

Method	Cofficient Prediction	HGCKs	mean $\mathbf{F_1}(\%)$
BSNet	✓	✓	87.6
		✓	86.9
	–	–	**85.6**

predictions around object boundaries and significantly improve segmentation accuracy.

2) Effect of Coefficient Prediction Module: To measure the performance gain from the CPM, we remove the CPM from the network and test its performance change. In other words, we directly fuse HGCKs into a convolution kernel. As shown in Table IV, after the CPM is removed, the mean F_1 score drops from 87.6% to 86.9%, proving the importance of the CPM.

3) Effect of DHGC and CSA: To verify the performance of DHGC and CSA, we conduct experiments with different settings in Table V. Compared with the baseline, the use of DHGC can achieve a mean F_1 score of 87.6%, corresponding to an increase of 3.3%. Meanwhile, after inserting CSA_1 into PSPNet with DHGC, the mean F_1 score reached 88.1%. When we integrate DHGC, CSA_1, and CSA_2 modules together, the performance is further improved to 88.9%. Fig. 16 displays some visualization results. DHGC can recover details and object boundaries more clearly, such as Car in the first line and Building in the second line. At the same time, more accurate boundaries can be obtained through our CSA.

TABLE V
ABLATION EXPERIMENT RESULTS ON DHGC AND CSA

Method	DHGC	CSA_1	CSA_2	mean $\mathbf{F_1}(\%)$
PSPNet (baseline)	–	–	–	84.3
BSNet	✓			87.6
	✓	✓		88.1
	✓	✓	✓	**88.9**

The integration of DHGC and CSA helps to restore fine segmentation boundaries. The segmentation quality of the remote sensing image has been clearly improved.

As we all know, vanilla convolutional kernels in shallow layers also can learn boundary information [100], [101]. However, the boundary information extracted is far from enough for remote sensing images with complex scenes. In order to verify the effectiveness of the boundary information extraction of our proposed DHGC, we sequentially replace the VCKs with DHGC in the Stages (1–5) of the baseline. As shown in Table VI, with the replacement of the VCKs with DHGC, the prediction performance of the network is gradually improved. When all VCKs are replaced, the mean F_1 performance and the F_1 of 5 and 7 px both reach best. As shown in Fig. 17, with the network deepening, the performance improvement brought by DHGC is waning. This also confirms that the network is capturing more detailed information at shallow layers.

Moreover, as shown in Fig. 18, we have listed the counterparts of CSA. We conducted related experiments based on the baseline. Table VII shows that our method achieves the best accuracy, which proves the effectiveness of our method. In order to verify the effectiveness of the preprocessing of DHGC, we compare the performance of NCA (as shown in Fig. 19, the NCA is constructed by replacing the DHGC of CSA with the 3×3 VC) and CSA, and the experiment proves that DHGC can greatly improve the performance compared with VC at similar inferencing speed. The comparison between LCA and CSA further proves the superiority of DHGC in accuracy and inference speed. In order to verify the effectiveness of LCA, we compare the performance of CA and LCA, and the experiment proves that LCA can not only accelerate the inference speed but also improve the accuracy accordingly.

C. Comparison With Encoder–Decoder Methods

This section verifies the superiority of BSNet through comparison to encoder–decoder segmentation models on the Vaihingen and Potsdam datasets. The comparison models include FCN-8s [25], UNet [47], SegNet [13], PSPNet [24], DeepLabv3+ [26], EncNet [97], S-RA-FCN [99], CCEM [55], and RefineNet [27]. For a fair comparison, ResNet18 is used as the backbone for PSPNet, EncNet, RefineNet, FCN-8s, CCEM, and S-RA-FCN. The IRRG is fed into these methods as the input.

1) Comparison on the Vaihingen Dataset: Table VIII summarizes the quantitative comparison of the proposed BSNet with state-of-the-art methods on the Vaihingen dataset.

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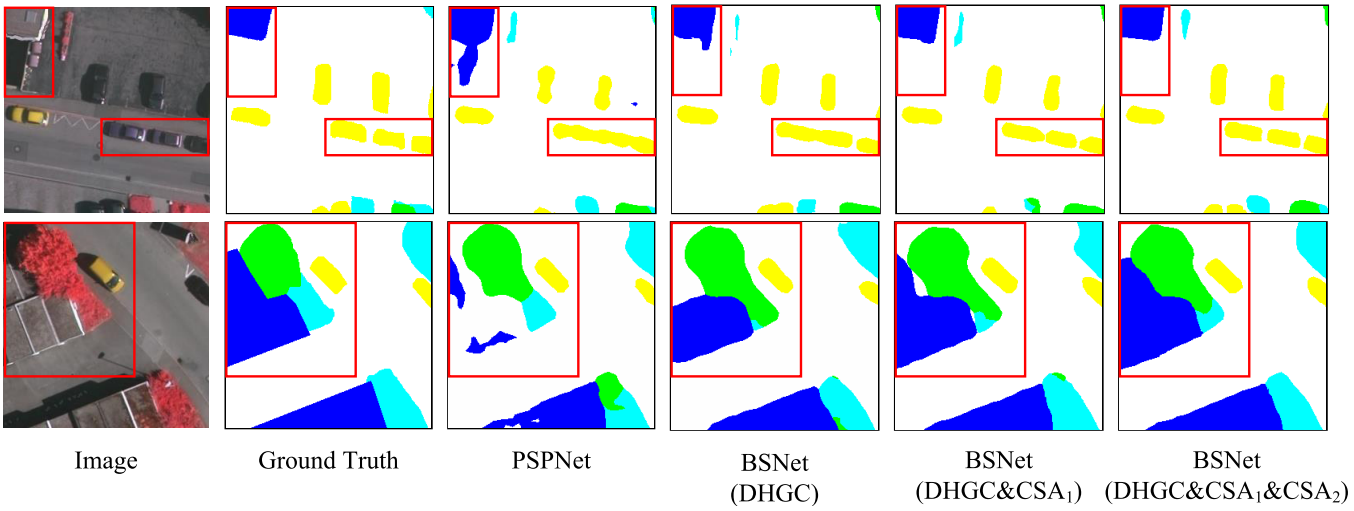


Fig. 16. Segmentation results when adopting different modules. The red boxes illustrate the improvements with DHGC and CSA.

TABLE VI
ABLATION EXPERIMENT RESULTS ON DHGC AT DIFFERENT STAGES

Method	Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	mean F_1(%)	5px F_1(%)	7px F_1(%)
BSNet						84.3	53.8	56.9
	✓					85.7	55.6	58.4
	✓	✓				86.5	56.6	59.3
	✓	✓	✓			87.0	57.3	59.9
	✓	✓	✓	✓		87.4	57.8	60.3
	✓	✓	✓	✓	✓	**87.6**	**58.0**	**60.6**

TABLE VII
COMPARATIVE EXPERIMENT RESULTS OF CSA AND ITS COUNTERPARTS

Method	mean F_1(%)	Average running time each image (second)
Baseline	84.3	0.017±0.001
Baseline+Skip Connection	85.5	0.019±0.001
Baseline+CA	86.9	0.030±0.001
Baseline+LCA	87.0	0.027±0.001
Baseline+NCA	86.7	**0.020±0.001**
Baseline+CSA	**87.4**	0.021±0.001

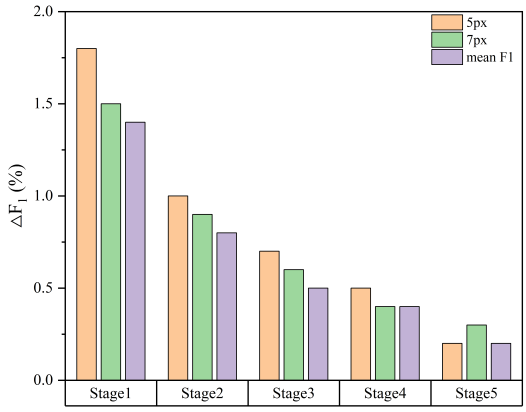


Fig. 17. Comparison of performance improvement on DHGC at different Stages.

As shown in Table VIII, the proposed BSNet provides segmentation results superior to other methods in terms of the OA and mean F_1.

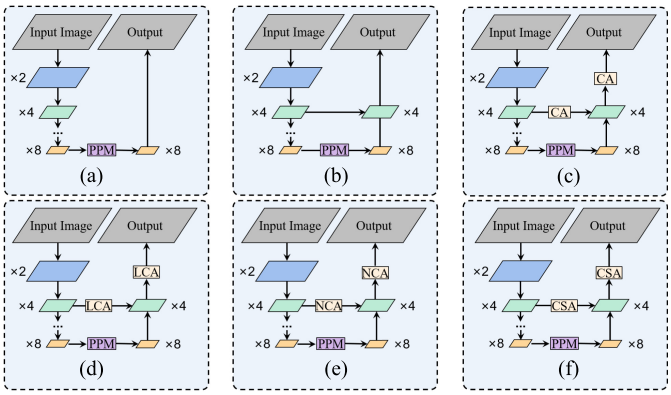


Fig. 18. Counterparts to compare against CSA. (a) Baseline. (b) Baseline + Skip connection. (c) Baseline + CA. (d) Baseline + LCA. (e) Baseline + NCA. (f) Baseline + CSA.

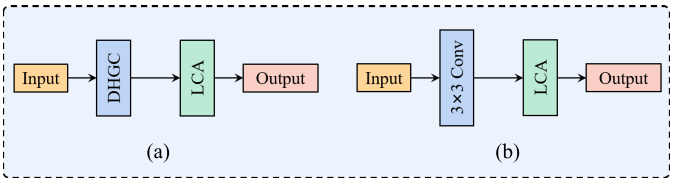


Fig. 19. Details of (a) CSA and (b) NCA.

The FCN segments large-scale Building well. However, its prediction of small objects, such as Tree and Car, is not satisfactory due to its lack of attention to detailed information. S-RA-FCN achieves higher accuracy than FCN since it can aggregate long-range spatial relations among the pixels. UNet

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TABLE VIII
COMPARISON WITH ENCODER–DECODER METHODS ON THE VAIHINGEN DATASET

Method	Impervious Surface	Building	Low Vegetation	Tree	Car	OA	mean F_1	mIoU
FCN-8s [11]	90.0	93.0	77.7	86.5	80.4	88.3	85.5	75.5
UNet [47]	90.5	93.3	79.6	87.5	76.4	89.2	85.5	75.5
SegNet [13]	90.2	93.7	78.5	85.8	83.9	88.5	86.4	76.8
EncNet [97]	91.2	94.1	79.2	86.9	83.7	89.4	87.0	77.8
RefineNet [98]	91.1	94.1	79.8	87.2	82.3	88.9	86.9	77.1
CCEM [55]	91.5	93.8	79.4	87.3	83.5	88.6	87.1	78.0
DeepLabv3+ [26]	91.4	**94.7**	79.6	87.6	85.8	89.9	87.8	79.0
S-RA-FCN [99]	90.5	93.8	79.6	87.5	82.6	89.2	86.8	77.3
PSPNet (baseline) [24]	90.6	94.3	79.0	87.0	70.7	89.1	84.3	74.1
BSNet	**92.1**	94.4	**83.1**	**88.3**	**86.7**	**90.3**	**88.9**	**80.2**

TABLE IX
COMPARISON WITH ENCODER–DECODER METHODS ON THE POTSDAM DATASET

Method	Impervious Surface	Building	Low Vegetation	Tree	Car	OA	mean F_1	mIoU
FCN-8s [11]	89.9	93.7	83.0	85.2	93.5	87.8	89.1	71.7
UNet [47]	88.2	91.1	82.8	84.9	91.6	86.2	87.7	68.5
SegNet [13]	87.8	90.7	81.0	84.7	89.7	85.7	86.8	66.2
EncNet [97]	91.0	94.9	84.4	85.9	93.6	89.0	90.0	73.2
RefineNet [98]	88.1	93.1	85.6	86.3	90.3	88.3	88.7	72.6
CCEM [55]	88.3	93.2	84.7	86.0	92.8	89.1	89.0	72.8
DeepLabv3+ [26]	91.3	94.8	84.2	86.6	93.8	89.2	90.1	73.8
S-RA-FCN [99]	90.7	94.2	83.8	85.8	93.6	88.5	89.6	72.5
PSPNet (baseline) [24]	90.7	94.8	84.1	85.9	90.5	88.7	89.2	71.7
BSNet	**92.4**	**95.6**	**86.8**	**88.1**	**94.6**	**90.7**	**91.5**	**77.5**

and SegNet improve the segmentation results by reusing the detailed features with skip connections. The low- and high-level features are combined to recover the details in UNet. The indices that record the position of the maximum in pooling are used in SegNet to upsample features. Furthermore, the chained residual pooling module adopted in RefineNet helps it exceed the segmentation results over UNet and SegNet. The module captures the background context to refine blurred segments, particularly at the boundaries of high-spatial-resolution images. CCEM is superior to RefineNet for Cars because it uses height maps as an additional channel to extract boundary information. In addition, Deeplabv3+ is superior to EncNet. This is because the context information captured by atrous convolutions can improve the boundary details.

The proposed BSNet obtains gains of 4.6%, 1.2%, and 6.1% in mean F_1 score, OA, and mIOU against PSPNet (baseline). The proposed method shows great improvement in the Low Vegetation and Car categories, particularly in the boundaries. Fig. 20 shows some examples of qualitative results. The results show that the segmentation boundaries of both small objects and large objects are significantly improved, which verifies the effectiveness of our method.

TABLE X
COMPARISON OF THE INFERENCE TIME ON EACH IMAGE WITH THE ENCODER–DECODER METHODS

Method	Average running time each image (second)
FCN-8s [11]	0.025±0.001
UNet [47]	0.039±0.001
SegNet [13]	0.081±0.001
EncNet [97]	0.019±0.001
RefineNet [98]	0.087±0.001
DeepLabv3+ [26]	0.091±0.001
PSPNet (baseline) [24]	0.017±0.001
BSNet	0.021±0.001

2) Comparison on the Potsdam Dataset: To further verify the superiority of BSNet, experiments are carried out on the Potsdam benchmark. As summarized in Table IX, the proposed BSNet demonstrated the optimal performance, exceeding the baseline by 2.3% in the mean F_1 score. Similar to the Vaihingen dataset, the proposed BSNet pro-

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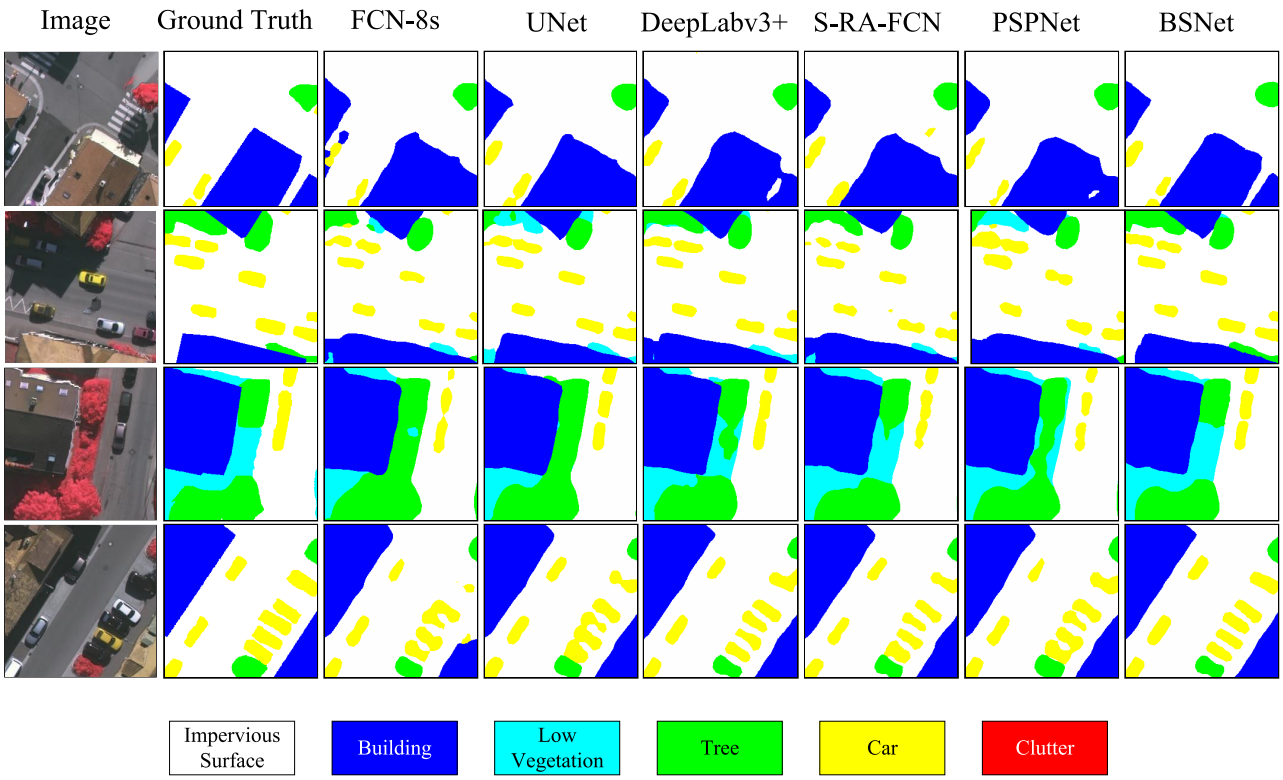


Fig. 20. Comparison of segmentation results (cropped images) between BSNet and encoder–decoder methods on Vaihingen.

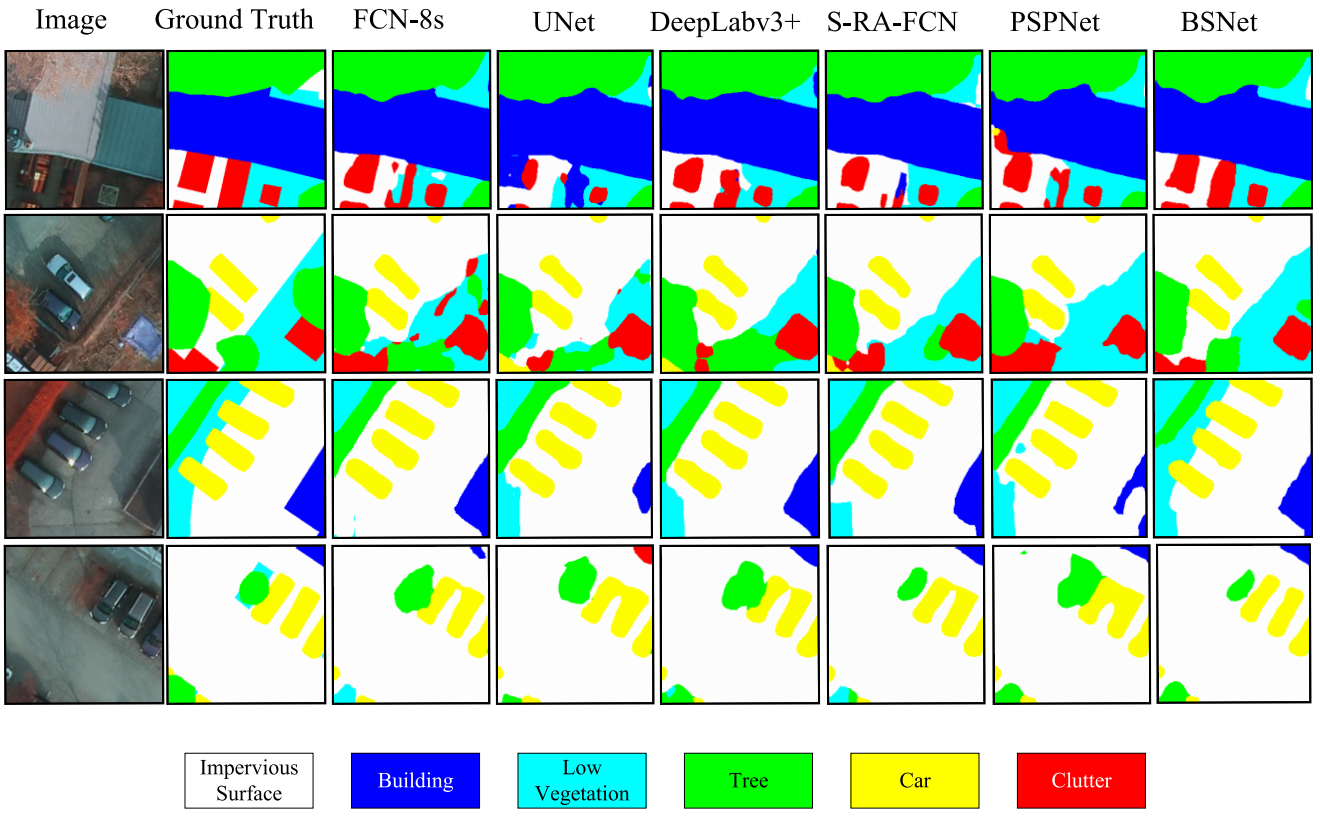


Fig. 21. Comparison of segmentation results (cropped images) between BSNet and encoder–decoder methods on Potsdam.

vides a sharper boundary and better accuracy. In particular, due to a large number of training samples in the Potsdam dataset, most comparison methods give better results on the Potsdam dataset than on the Vaihingen dataset. Several segmentation results on the Potsdam dataset are shown in Fig. 21.

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TABLE XI
COMPARISON WITH HIGH-QUALITY METHODS ON THE VAIHINGEN DATASET

Method	Backbone	OA	mean F_1	mIoU
UZ_1 [102]	–	87.3	81.5	–
UFMG_4 [49]	–	89.4	87.7	–
RoteEqnet [103]	–	87.5	84.2	–
CVEO [104]	–	88.3	86.5	–
CAM-DFCN [105]	ResNet-101	89.0	87.1	80.9
CASIA2 [106]	ResNet-101	**91.1**	90.1	–
CCNet [91]	ResNet-101	**91.1**	90.3	82.6
SBANet [41]	ResNet-101	90.6	90.3	–
BKHN11	ResNet-101	91.0	90.4	–
ACFNet [107]	ResNet-101	90.9	90.3	82.7
PSPNet (baseline) [24]	ResNet-101	90.1	88.7	81.2
BSNet	ResNet-101	90.5	**90.6**	**82.8**

TABLE XII
COMPARISON WITH HIGH-QUALITY METHODS ON THE POTSDAM DATASET

Method	Backbone	OA	mean F_1	mIoU
UZ_1 [102]	–	85.8	86.7	–
UFMG_4 [49]	–	87.9	89.5	–
AMA_1	–	91.2	92.5	–
HUSTW4 [108]	–	91.6	92.6	–
V-FuseNet [109]	–	90.6	92.0	–
CASIA3 [110]	ResNet-101	91.0	92.4	–
BKHN3	ResNet-101	91.1	92.6	–
CCNet [91]	ResNet-101	91.5	92.4	85.7
SBANet [41]	ResNet-101	92.8	93.0	–
SWJ_2	ResNet-101	91.7	92.4	–
PSPNet (baseline) [24]	ResNet-101	91.1	91.5	84.2
BSNet	ResNet-101	**92.9**	**93.1**	**86.9**

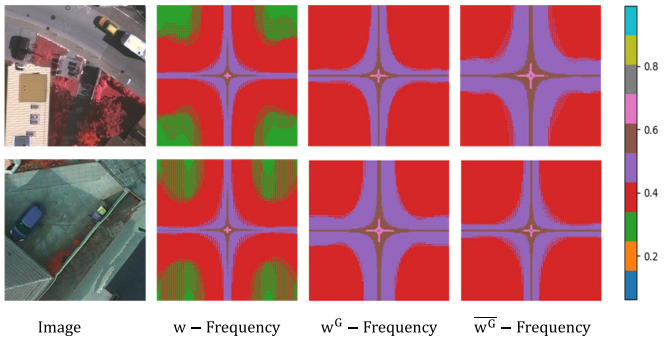


Fig. 22. Frequency analysis for feature maps from different kernels in Stage 2. The first and second rows show examples from Vaihingen and Potsdam datasets, respectively. Columns show the input images, the frequency maps of VCK ($\mathbf{w}$), gradient kernel ($\mathbf{w^G}$), and gradient kernel ($\overline{\mathbf{w^G}}$). In the frequency maps, the center point corresponds to the lowest frequency (dc component), and the frequency gradually increases from the center to the periphery.

D. Frequency Analysis and Qualitative Results on Vaihingen and Potsdam Datasets

The boundary information with sharp gradient changes corresponds to high-frequency components in the frequency domain. Fig. 22 shows the frequency analysis results. We apply the Fourier transform in feature maps of each kernel from Stage 2 and visualize the results. Compared with VCK, GCKs have a stronger response at the four corners, which corresponds to more high-frequency information being captured. This is due to the explicit boundary information extraction capability of GCKs. In addition, from the center point to the corner, $\mathbf{w^G}$ and $\overline{\mathbf{w^G}}$ have different degrees of enhanced ability to extract high-frequency information. This is because the gradient distribution of boundary information in different images varies. Therefore, using $\mathbf{w^G}$ and $\overline{\mathbf{w^G}}$ can capture richer boundary information, which justifies the necessity of these two kernels.

Fig. 23 visualizes the prediction errors between BSNet and the baseline. It can be observed that segmentation errors are mainly distributed within the boundaries. Compared with PSPNet, our method significantly improves the segmentation quality along the boundaries. The error boundaries for BSNet become more narrow.

E. Evaluation of Inference Speed

To compare the inference speed, we compare the inference time of different models under the same input (256 ×

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TABLE XIII
RESULT COMPARISONS WITH STATE OF THE ARTS ON THE iSAID VALIDATION SET

Method	Backbone	IoU per category(%)															mIoU(%)	mean F_1	Acc.(%)
		PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC			
EMANet [111]	ResNet-50	77.8	71.2	28.3	53.1	40.0	56.9	58.4	82.6	52.4	62.8	63.5	40.2	48.7	43.1	30.8	56.8	70.3	94.0
DenseASPP [112]	ResNet-50	78.2	66.9	28.4	**53.5**	38.2	57.4	60.8	82.4	53.7	50.1	69.9	54.8	51.2	43.4	25.7	57.1	70.6	94.2
Non-local [88]	ResNet-50	81.0	71.9	29.1	49.8	47.8	59.1	62.4	84.9	48.6	62.2	41.6	53.5	50.9	45.1	29.9	57.3	70.8	94.3
DANet [113]	ResNet-50	80.9	71.4	28.6	52.5	42.1	57.5	60.2	84.7	50.9	63.0	63.3	40.6	48.8	46.1	30.4	57.5	71.0	94.4
CCNet [91]	ResNet-50	78.8	72.4	30.5	52.2	44.4	53.1	56.3	85.3	49.2	63.2	67.1	51.8	48.7	43.6	29.2	57.8	71.3	94.6
Semantic FPN [115]	ResNet-50	81.2	71.5	33.8	52.2	45.4	60.1	63.5	**87.1**	57.8	61.5	60.2	59.0	51.5	46.6	31.2	60.1	73.5	94.8
DeepLabv3+ [25]	ResNet-50	82.9	71.4	32.2	52.0	48.5	60.1	64.5	81.1	55.7	68.5	61.0	60.2	53.0	47.1	30.8	60.5	73.9	94.8
SETR [116]	T-Large	81.5	74.5	30.8	52.3	48.8	58.6	63.4	86.9	57.4	67.9	68.0	61.6	52.9	46.2	32.6	61.4	74.4	95.3
UperNet [117]	Swin-B	**83.0**	75.3	30.5	51.8	**49.6**	59.2	63.1	85.7	**59.2**	68.2	68.1	59.5	53.3	46.8	**34.3**	61.6	74.7	95.5
PSPNet (baseline) [24]	ResNet-50	82.8	**76.8**	31.2	53.4	47.3	58.2	63.9	86.5	56.7	68.3	67.4	52.6	53.0	44.0	33.3	60.9	74.2	95.1
BSNet	ResNet-50	81.8	76.1	**41.3**	50.3	46.6	**63.4**	**65.3**	86.8	55.9	**69.1**	**70.2**	**70.0**	**57.3**	**48.8**	31.8	**63.4**	**76.6**	**96.4**

* iSAID includes 15 foreground categories, *i.e.*, plane (PL), baseball diamond (BD), bridge (BR), ground track field (GTF), small vehicle (SV), large vehicle (LV), ship (SH), tennis court (TC), baseball court (BC), storage tank (ST), soccer ball field (SBF), roundabout (RA), harbor (HA), swimming pool (SP) and helicopter (HC).

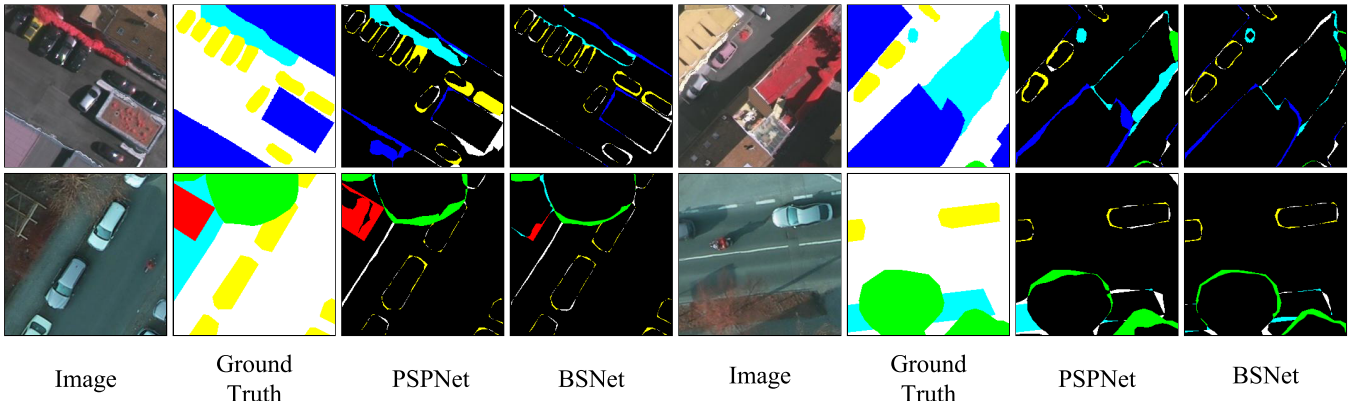


Fig. 23. Qualitative comparison of prediction errors. The black area represents the correct prediction result. Compared with the baseline, our method produces more precise boundaries.

256 pixels). The inference time is the average time obtained across all test samples, and each model is run five times to obtain a fair result. All inference time is tested on the same PC equipped with Intel Xeon 2.10-GHz CPU, 16-GB RAM, and an RTX 2080TI GPU. As shown in Table X, the computational cost of our method compared to the baseline is almost negligible. Compared to EncNet, our model has a better tradeoff between accuracy and speed. In terms of accuracy and speed, our model shows a significant advantage over FCN-8s, UNet, SegNet, RefineNet, and DeepLabv3+. The above results favorably prove the superiority of the BSNet.

F. Comparison With High-Quality Methods

All the abovementioned BSNets are based on ResNet-18. To further verify the performance of the proposed BSNet, we compare BSNet with several high-quality benchmark models, including UZ_1 [102], UFMG_4 [51], RoteEqnet [103], CVEO [104], CAM-DFCN [105], CASIA2 [106], CASIA3 [110], CCNet [91], BKHN3, BKHN11, ACFNet [107], and SWJ_2. For a fair comparison, we use ResNet-101 as the backbone.

As can be seen in Tables XI and XII, the mIoU and mean F_1 scores obtained by BSNet are higher than those of other benchmark models. In addition, compared with PSPNet (baseline), BSNet has a greater performance improvement. The results fully demonstrate the superiority of our method.

G. Experiments on iSAID Dataset

We further validate the effectiveness of the proposed method on the more challenging iSAID dataset. To the best of our knowledge, iSAID is currently the largest remote sensing dataset with the foreground–background imbalance problem.

For a fair comparison, we use ResNet-50 as the backbone for all models and choose the PSPNet (ResNet-50) as the baseline. All models are trained using only finely annotated 1411 images and validated on the predefined validation set (458 images). The numerical results on the validation set are shown in Table XIII. It can be seen that our approach achieves the best 63.4% mIoU, which outperforms the baseline by a large margin. It also surpasses the well verified methods (EMANet [111],

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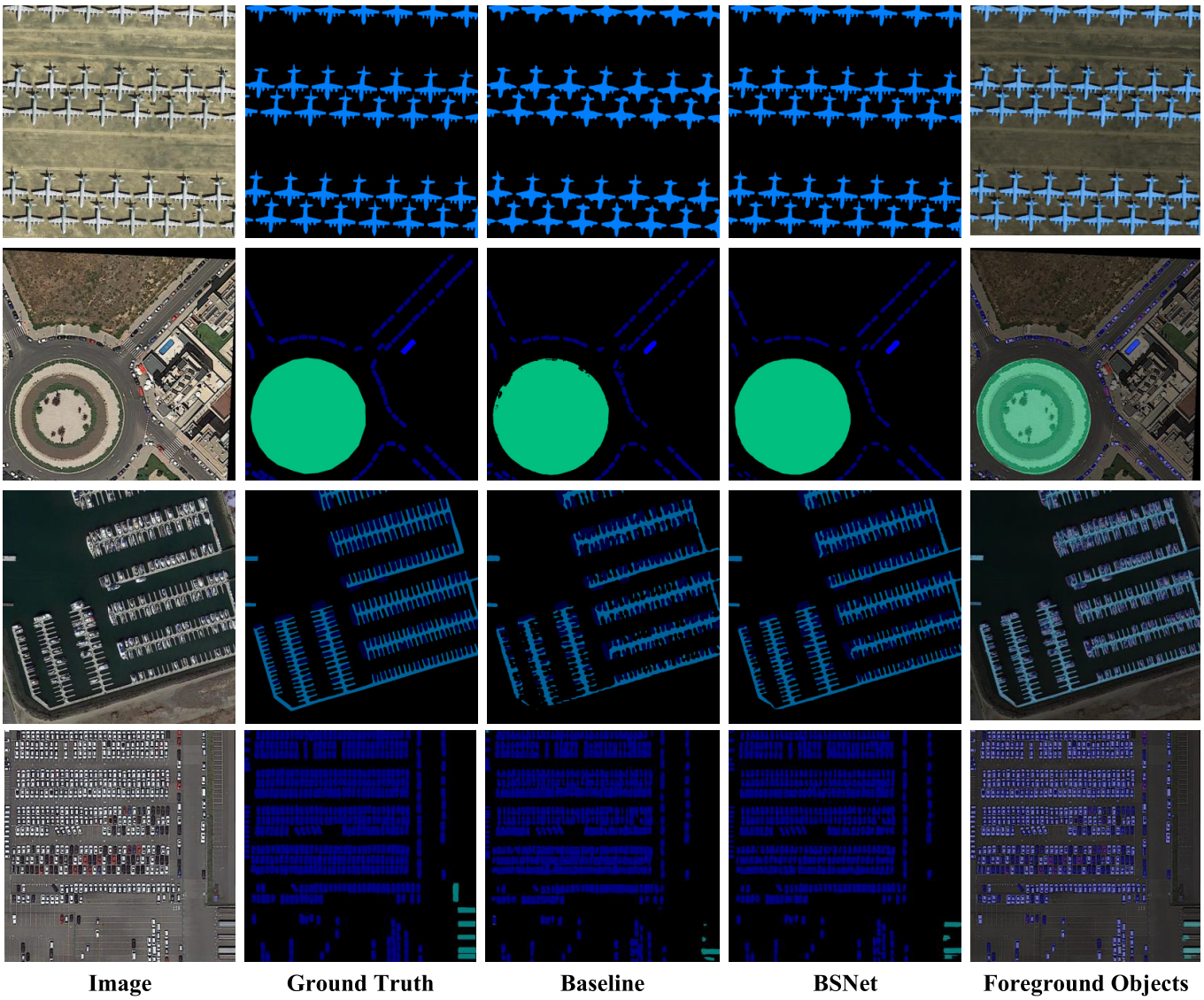


Fig. 24. Qualitative comparison between BSNet (ResNet50) and baseline (PSPNet + ResNet50) on the iSAID validation set.

DenseASPP [112], Non-local [88], DANet [113], CCNet [91], Semantic FPN [114], DeepLabv3+ [25], SETR [115], UperNet [116], and so on).

Some visualized results of BSNet (ResNet-50) on the iSAID validation set are shown in Fig. 24. The last column visualizes the segmentation results of the foreground objects. Compared with the baseline, our model has a better perception of foreground objects and can predict precise and thin boundary features.

V. Conclusion

This article proposes the BSNet to recover complete and accurate boundaries in the semantic segmentation of remote sensing images. BSNet can refine the boundaries in two ways. In the feature extraction stage, BSNet uses DHGC to enhance the boundary information in the semantic segmentation information flow. In the feature recovery stage, the CSA module adaptively mines boundary information and long-range dependencies in low-level features to help high-level features recover more precise boundaries. Experiments show that BSNet is an efficient architecture that can produce clearer predictions of object boundaries. The proposed BSNet achieves better performance than several prevalent methods on three challenging datasets: Vaihingen, Potsdam, and iSAID. In the future, we plan to focus on reducing the number of parameters and further improving the inference speed of the network.

References

[1] L. Lopez-Fuentes, C. Rossi, and H. Skinnemoen, "River segmentation for flood monitoring," in *Proc. IEEE Int. Conf. Big Data (Big Data)*, J. Nie *et al.*, Eds. Boston, MA, USA: IEEE Computer Society, Dec. 2017, pp. 3746–3749, doi: 10.1109/BigData.2017.8258373.

[2] K. Nogueira *et al.*, "Exploiting convnet diversity for flooding identification," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 9, pp. 1446–1450, Sep. 2018, doi: 10.1109/LGRS.2018.2845549.

[3] H. Sheng, X. Chen, J. Su, R. Rajagopal, and A. Ng, "Effective data fusion with generalized vegetation index: Evidence from land cover segmentation in agriculture," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Seattle, WA, USA, Jun. 2020, pp. 267–276, doi: 10.1109/CVPRW50498.2020.00038.

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[4] L. Mi and Z. Chen, "Superpixel-enhanced deep neural forest for remote sensing image semantic segmentation," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 140–152, 2020.

[5] M. M. Nielsen, "Remote sensing for urban planning and management: The use of window-independent context segmentation to extract urban features in Stockholm," *Comput., Environ. Urban Syst.*, vol. 52, pp. 1–9, Jul. 2015, doi: 10.1016/j.compenvurbsys.2015.02.002.

[6] Y. Liu, L. Gross, Z. Li, X. Li, X. Fan, and W. Qi, "Automatic building extraction on high-resolution remote sensing imagery using deep convolutional encoder-decoder with spatial pyramid pooling," *IEEE Access*, vol. 7, pp. 128774–128786, 2019, doi: 10.1109/ACCESS.2019.2940527.

[7] M. Fauvel, Y. Tarabalka, J. A. Benediktsson, J. Chanussot, and J. C. Tilton, "Advances in spectral-spatial classification of hyperspectral images," *Proc. IEEE*, vol. 101, no. 3, pp. 652–675, Mar. 2013, doi: 10.1109/JPROC.2012.2197589.

[8] L. Gueguen, "Classifying compound structures in satellite images: A compressed representation for fast queries," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 4, pp. 1803–1818, Apr. 2015, doi: 10.1109/TGRS.2014.2348864.

[9] F. Li, X. Jia, D. Fraser, and A. Lambert, "Super resolution for remote sensing images based on a universal hidden Markov tree model," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 3, pp. 1270–1278, Mar. 2010, doi: 10.1109/TGRS.2009.2031636.

[10] F. Tupin and M. Roux, "Markov random field on region adjacency graph for the fusion of SAR and optical data in radargrammetric applications," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 8, pp. 1920–1928, Aug. 2005, doi: 10.1109/TGRS.2005.852080.

[11] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440, doi: 10.1109/CVPR.2015.7298965.

[12] H. Noh, S. Hong, and B. Han, "Learning deconvolution network for semantic segmentation," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1520–1528, doi: 10.1109/ICCV.2015.178.

[13] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder-decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Jan. 2017, doi: 10.1109/TPAMI.2016.2644615.

[14] A. Li, L. Jiao, H. Zhu, L. Li, and F. Liu, "Multitask semantic boundary awareness network for remote sensing image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022, doi: 10.1109/TGRS.2021.3050885.

[15] S. Liu, W. Ding, C. Liu, Y. Liu, Y. Wang, and H. Li, "ERN: Edge loss reinforced semantic segmentation network for remote sensing images," *Remote Sens.*, vol. 10, no. 9, p. 1339, Aug. 2018, doi: 10.3390/rs10091339.

[16] G. Cheng, X. Xie, J. Han, L. Guo, and G.-S. Xia, "Remote sensing image scene classification meets deep learning: Challenges, methods, benchmarks, and opportunities," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, no. 99, pp. 3735–3756, Jun. 2020, doi: 10.1109/JSTARS.2020.3005403.

[17] G. Cheng, C. Yang, X. Yao, L. Guo, and J. Han, "When deep learning meets metric learning: Remote sensing image scene classification via learning discriminative CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 5, pp. 2811–2821, May 2018, doi: 10.1109/TGRS.2017.2783902.

[18] Y. Li, T. Shi, Y. Zhang, W. Chen, Z. Wang, and H. Li, "Learning deep semantic segmentation network under multiple weakly-supervised constraints for cross-domain remote sensing image semantic segmentation," *ISPRS J. Photogramm. Remote Sens.*, vol. 175, pp. 20–33, May 2021.

[19] Y. Li, W. Chen, Y. Zhang, C. Tao, R. Xiao, and Y. Tan, "Accurate cloud detection in high-resolution remote sensing imagery by weakly supervised deep learning," *Remote Sens. Environ.*, vol. 250, Dec. 2020, Art. no. 112045.

[20] T. Takikawa, D. Acuna, V. Jampani, and S. Fidler, "Gated-SCNN: Gated shape CNNs for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Nov. 2019, pp. 5228–5237, doi: 10.1109/ICCV.2019.00533.

[21] L.-C. Chen, J. T. Barron, G. Papandreou, K. Murphy, and A. L. Yuille, "Semantic image segmentation with task-specific edge detection using CNNs and a discriminatively trained domain transform," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 4545–4554, doi: 10.1109/CVPR.2016.492.

[22] K. Gong, X. Liang, Y. Li, Y. Chen, M. Yang, and L. Lin, "Instance-level human parsing via part grouping network," *CoRR*, vol. abs/1808.00157, 2018. [Online]. Available: http://arxiv.org/abs/1808.00157

[23] X. Li *et al.*, "Semantic flow for fast and accurate scene parsing," *CoRR*, vol. abs/2002.10120, 2020. [Online]. Available: https://arxiv.org/abs/2002.10120

[24] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Honolulu, HI, USA, Jul. 2017, pp. 6230–6239, doi: 10.1109/CVPR.2017.660.

[25] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.

[26] L. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder-decoder with atrous separable convolution for semantic image segmentation," *CoRR*, vol. abs/1802.02611, 2018. [Online]. Available: http://arxiv.org/abs/1802.02611

[27] Y. Wang, W. Ding, R. Zhang, and H. Li, "Boundaryaware multitask learning for remote sensing imagery," *IEEE J. Sel. Topics. Appl. Earth Observ. Remote. Sens.*, vol. 14, pp. 951–963, 2021, doi: 10.1109/JSTARS.2020.3043442.

[28] G. Yang, Q. Zhang, and G. Zhang, "EANet: Edge-aware network for the extraction of buildings from aerial images," *Remote Sens.*, vol. 12, no. 13, p. 2161, Jul. 2020, doi: 10.3390/rs12132161.

[29] X. Sun, A. Shi, H. Huang, and H. Mayer, "BAS^{4}Net: Boundary-aware semi-supervised semantic segmentation network for very high resolution remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 5398–5413, 2020, doi: 10.1109/JSTARS.2020.3021098.

[30] Z. Su *et al.*, "Pixel difference networks for efficient edge detection," 2021, *arXiv:2108.07009*.

[31] I. Sobel and G. Feldman, "A 3×3 isotropic gradient operator for image processing," in *A Talk at the Stanford Artificial Project in*. 1968, pp. 271–272.

[32] J. M. Prewitt, "Object enhancement and extraction," *Picture Process. Psychopictorics*, vol. 10, no. 1, pp. 15–19, 1970.

[33] J. Canny, "A computational approach to edge detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. PAMI-8, no. 6, pp. 679–698, Nov. 1986.

[34] P. Dollár and C. L. Zitnick, "Fast edge detection using structured forests," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 8, pp. 1558–1570, Aug. 2015.

[35] S. Hallman and C. C. Fowlkes, "Oriented edge forests for boundary detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 1732–1740.

[36] D. R. Martin, C. C. Fowlkes, and J. Malik, "Learning to detect natural image boundaries using local brightness, color, and texture cues," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 26, no. 5, pp. 530–549, 2004, doi: 10.1109/TPAMI.2004.1273918.

[37] P. Dollár and C. L. Zitnick, "Fast edge detection using structured forests," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 8, pp. 1558–1570, 2015, doi: 10.1109/TPAMI.2014.2377715.

[38] D. R. Martin, C. C. Fowlkes, and J. Malik, "Learning to detect natural image boundaries using local brightness, color, and texture cues," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 26, no. 5, pp. 530–549, May 2004.

[39] X. Ren and L. Bo, "Discriminatively trained sparse code gradients for contour detection," in *Proc. 25th Int. Conf. Neural Inf. Process. Syst.*, vol. 1, 2012, pp. 584–592.

[40] R. C. Gonzalez and R. E. Woods, *Digital Image Processing*. Reading, MA, USA: Addison-Wesley, 1977.

[41] Q. Hou, D. Zhou, and J. Feng, "Coordinate attention for efficient mobile network design," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 13713–13722. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2021/html/Hou_Coordinate_Attention_for_Efficient_Mobile_Network_Design_CVPR_2021_paper.html

[42] R. Achanta *et al.*, "SLIC superpixels compared to state-of-the-art superpixel methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 11, pp. 2274–2282, May 2012, doi: 10.1109/TPAMI.2012.120.

[43] T.-W. Chen, Y.-L. Chen, and S.-Y. Chien, "Fast image segmentation based on K-means clustering with histograms in HSV color space," in *Proc. IEEE 10th Workshop Multimedia Signal Process.*, Oct. 2008, pp. 322–325.

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[44] G. Fu, C. Liu, R. Zhou, T. Sun, and Q. Zhang, "Classification for high resolution remote sensing imagery using a fully convolutional network," *Remote Sens.*, vol. 9, no. 6, p. 498, 2017, doi: 10.3390/rs9050498.

[45] Y. Liu, B. Fan, L. Wang, J. Bai, S. Xiang, and C. Pan, "Semantic labeling in very high resolution images via a self-cascaded convolutional neural network," *ISPRS J. Photogram. Remote Sens.*, vol. 145, pp. 78–95, 2018, doi: 10.1016/j.isprsjprs.2017.12.007.

[46] Y. Feng, X. Sun, W. Diao, J. Li, X. Gao, and K. Fu, "Continual learning with structured inheritance for semantic segmentation in aerial imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–17, 2022.

[47] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," *CoRR*, vol. abs/1505.04597, 2015. [Online]. Available: http://arxiv.org/abs/1505.04597

[48] Z. Zhou, M. M. R. Siddiquee, N. Tajbakhsh, and J. Liang, "UNet++: Redesigning skip connections to exploit multiscale features in image segmentation," *IEEE Trans. Med. Imag.*, vol. 39, no. 6, pp. 1856–1867, Jun. 2020, doi: 10.1109/TMI.2019.2959609.

[49] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018, doi: 10.1109/TPAMI.2017.2699184.

[50] R. Hamaguchi, A. Fujita, K. Nemoto, T. Imaizumi, and S. Hikosaka, "Effective use of dilated convolutions for segmenting small object instances in remote sensing imagery," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Mar. 2018, pp. 1442–1450, doi: 10.1109/WACV.2018.00162.

[51] K. Nogueira, M. D. Mura, J. Chanussot, W. R. Schwartz, and J. A. dos Santos, "Dynamic multicontext segmentation of remote sensing images based on convolutional networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7503–7520, Oct. 2019, doi: 10.1109/TGRS.2019.2913861.

[52] F. Yu and V. Koltun, "Multi-scale context aggregation by dilated convolutions," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, May 2016.

[53] P. Wang *et al.*, "Understanding convolution for semantic segmentation," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Lake Tahoe, NV, USA, Mar. 2018, pp. 1451–1460, doi: 10.1109/WACV.2018.00163.

[54] A. Ghosh, M. Ehrlich, S. Shah, L. Davis, and R. Chellappa, "Stacked U-Nets for ground material segmentation in remote sensing imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 252–2524, doi: 10.1109/CVPRW.2018.00047.

[55] D. Marmanis, K. Schindler, J. D. Wegner, S. Galliani, M. Datcu, and U. Stilla, "Classification with an edge: Improving semantic image segmentation with boundary detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 135, pp. 158–172, Jan. 2018, doi: 10.1016/j.isprsjprs.2017.11.009.

[56] H. Wang, Y. Wang, Q. Zhang, S. Xiang, and C. Pan, "Gated convolutional neural network for semantic segmentation in high-resolution images," *Remote Sens.*, vol. 9, no. 5, p. 446, 2017, doi: 10.3390/rs9050446.

[57] L. Mou, Y. Hua, and X. X. Zhu, "Relation matters: Relational context-aware fully convolutional network for semantic segmentation of high-resolution aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 7557–7569, Nov. 2020, doi: 10.1109/TGRS.2020.2979552.

[58] F. Zhou, R. Hang, and Q. Liu, "Class-guided feature decoupling network for airborne image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2245–2255, Mar. 2021, doi: 10.1109/TGRS.2020.3006872.

[59] N. He, L. Fang, and A. Plaza, "Hybrid first and second order attention Unet for building segmentation in remote sensing images," *Sci. China Inf. Sci.*, vol. 63, no. 4, Mar. 2020, Art. no. 140305, doi: 10.1007/s11432-019-2791-7.

[60] Z. Yu, C. Feng, M.-Y. Liu, and S. Ramalingam, "CASENet: Deep category-aware semantic edge detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1761–1770, doi: 10.1109/CVPR.2017.191.

[61] T. Hu, Y. Wang, Y. Chen, P. Lu, H. Wang, and G. Wang, "Sobel heuristic kernel for aerial semantic segmentation," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Athens, Greece, Oct. 2018, pp. 3074–3078, doi: 10.1109/ICIP.2018.8451170.

[62] S. Xie and Z. Tu, "Holistically-nested edge detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1395–1403, doi: 10.1109/ICCV.2015.164.

[63] G. Bertasius, J. Shi, and L. Torresani, "High-for-low and low-for-high: Efficient boundary detection from deep object features and its applications to high-level vision," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 504–512, doi: 10.1109/ICCV.2015.65.

[64] J. Chen, F. He, Y. Zhang, G. Sun, and M. Deng, "SPMF-Net: Weakly supervised building segmentation by combining superpixel pooling and multi-scale feature fusion," *Remote Sens.*, vol. 12, no. 6, p. 1049, Mar. 2020, doi: 10.3390/rs12061049.

[65] D. Cheng, G. Meng, S. Xiang, and C. Pan, "Fusionnet: Edge aware deep convolutional networks for semantic segmentation of remote sensing harbor images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 12, pp. 5769–5783, Dec. 2017, doi: 10.1109/JSTARS.2017.2747599.

[66] G. Bertasius, J. Shi, and L. Torresani, "Semantic segmentation with boundary neural fields," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 3602–3610, doi: 10.1109/CVPR.2016.392.

[67] Y. Chong, X. Chen, and S. Pan, "Context union edge network for semantic segmentation of small-scale objects in very high resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2020.3021210.

[68] X. Li *et al.*, "Improving semantic segmentation via decoupled body and edge supervision," in *Proc. 16th Eur. Conf. Comput. Vis.* (Lecture Notes in Computer Science), vol. 12362, A. Vedaldi, H. Bischof, T. Brox, and J. Frahm, Eds. Glasgow, U.K.: Springer, 2020, pp. 435–452, doi: 10.1007/978-3-030-58520-4_26.

[69] B. Klein, L. Wolf, and Y. Afek, "A dynamic convolutional layer for short rangeweather prediction," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 4840–4848, doi: 10.1109/CVPR.2015.7299117.

[70] B. D. Brabandere, X. Jia, T. Tuytelaars, and L. V. Gool, "Dynamic filter networks," *CoRR*, vol. abs/1605.09673, 2016. [Online]. Available: http://arxiv.org/abs/1605.09673

[71] D. Shen, M. R. Min, Y. Li, and L. Carin, "Learning context-sensitive convolutional filters for text processing," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2018, pp. 1839–1848, doi: 10.18653/v1/D18-1210.

[72] F. Wu, A. Fan, A. Baevski, Y. N. Dauphin, and M. Auli, "Pay less attention with lightweight and dynamic convolutions," *CoRR*, vol. abs/1901.10430, 2019. [Online]. Available: http://arxiv.org/abs/1901.10430

[73] J. Gong, X. Qiu, X. Chen, D. Liang, and X. Huang, "Convolutional interaction network for natural language inference," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2018, pp. 1576–1585, doi: 10.18653/v1/D18-1186.

[74] B. Yang, G. Bender, Q. V. Le, and J. Ngiam, "Condconv: Conditionally parameterized convolutions for efficient inference," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019.

[75] Y. Chen, X. Dai, M. Liu, D. Chen, L. Yuan, and Z. Liu, "Dynamic convolution: Attention over convolution kernels," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11027–11036, doi: 10.1109/CVPR42600.2020.01104.

[76] J. Chen, X. Wang, Z. Guo, X. Zhang, and J. Sun, "Dynamic region-aware convolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 8064–8073.

[77] Y. Zhang, J. Zhang, Q. Wang, and Z. Zhong, "DyNet: Dynamic convolution for accelerating convolutional neural networks," 2020, *arXiv:2004.10694*.

[78] M. Xu, R. Ding, H. Zhao, and X. Qi, "PAConv: Position adaptive convolution with dynamic kernel assembling on point clouds," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 3173–3182. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2021/html/Xu_PAConv_Position_Adaptive_Convolution_With_Dynamic_Kernel_Assembling_on_Point_CVPR_2021_paper.html

[79] A. Vaswani *et al.*, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017.

[80] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141, doi: 10.1109/CVPR.2018.00745.

[81] S. Woo, J. Park, J. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," *CoRR*, vol. abs/1807.06521, 2018. [Online]. Available: http://arxiv.org/abs/1807.06521

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[82] J. Hu, L. Shen, S. Albanie, G. Sun, and A. Vedaldi, "Gatherexcite: Exploiting feature context in convolutional neural networks," *Proc. Adv. Neural Inf. Process. Syst.*, vol. 31, 2018.

[83] D. Linsley, D. Shiebler, S. Eberhardt, and T. Serre, "Learning what and where to attend," 2018, *arXiv:1805.08819*.

[84] I. Bello, B. Zoph, Q. Le, A. Vaswani, and J. Shlens, "Attention augmented convolutional networks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Nov. 2019, pp. 3285–3294, doi: 10.1109/ICCV.2019.00338.

[85] H. Li, P. Xiong, J. An, and L. Wang, "Pyramid attention network for semantic segmentation," in *Proc. Brit. Mach. Vis. Conf.* Newcastle, U.K.: BMVA Press, Sep. 2018, p. 285. [Online]. Available: http://bmvc2018.org/contents/papers/1120.pdf

[86] Y. Cao, J. Xu, S. Lin, F. Wei, and H. Hu, "GCNet: Non-local networks meet squeeze-excitation networks and beyond," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Oct. 2019, pp. 1971–1980, doi: 10.1109/ICCVW.2019.00246.

[87] Y. Chen, Y. Kalantidis, J. Li, S. Yan, and J. Feng, "A^2-Nets: Double attention networks," in *Proc. Adv. Neural Inf. Process. Syst., Annu. Conf. Neural Inf. Process. Syst. (NeurIPS)*, S. Bengio, H. M. Wallach, H. Larochelle, K. Grauman, N. Cesa-Bianchi, and R. Garnett, Eds. Montreal, QC, Canada, 2018, pp. 350–359. [Online]. Available: https://proceedings.neurips.cc/paper/2018/hash/e165421110ba03099a1c0393373c5b43-Abstract.html

[88] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7794–7803, doi: 10.1109/CVPR.2018.00813.

[89] J.-J. Liu, Q. Hou, M.-M. Cheng, C. Wang, and J. Feng, "Improving convolutional networks with self-calibrated convolutions," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10093–10102, doi: 10.1109/CVPR42600.2020.01011.

[90] Z. Gao, J. Xie, Q. Wang, and P. Li, "Global second-order pooling convolutional networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3019–3028, doi: 10.1109/CVPR.2019.00314.

[91] Z. Huang, X. Wang, L. Huang, C. Huang, Y. Wei, and W. Liu, "CCNet: Criss-cross attention for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Nov. 2019, pp. 603–612, doi: 10.1109/ICCV.2019.00069.

[92] Q. Hou, L. Zhang, M.-M. Cheng, and J. Feng, "Strip pooling: Rethinking spatial pooling for scene parsing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Seattle, WA, USA, Jun. 2020, pp. 4002–4011. [Online]. Available: https://openaccess.thecvf.com/content_CVPR_2020/html/Hou_Strip_Pooling_Rethinking_Spatial_Pooling_for_Scene_Parsing_CVPR_2020_paper.html

[93] C. Peng, K. Zhang, Y. Ma, and J. Ma, "Cross fusion net: A fast semantic segmentation network for small-scale semantic information capturing in aerial scenes," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022, doi: 10.1109/TGRS.2021.3053062.

[94] Z. Zheng, Y. Zhong, J. Wang, and A. Ma, "Foreground-aware relation network for geospatial object segmentation in high spatial resolution remote sensing imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Seattle, WA, USA, Jun. 2020, pp. 4095–4104. [Online]. Available: https://openaccess.thecvf.com/content_CVPR_2020/html/Zheng_Foreground-Aware_Relation_Network_for_Geospatial_Object_Segmentation_in_High_Spatial_CVPR_2020_paper.html

[95] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 248–255, doi: 10.1109/CVPR.2009.5206848.

[96] J. Xie, N. He, L. Fang, and A. Plaza, "Scale-free convolutional neural network for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6916–6928, Sep. 2019, doi: 10.1109/TGRS.2019.2909695.

[97] H. Zhang *et al.*, "Context encoding for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7151–7160, doi: 10.1109/CVPR.2018.00747.

[98] G. Lin, A. Milan, C. Shen, and I. Reid, "RefineNet: Multi-path refinement networks for high-resolution semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5168–5177, doi: 10.1109/CVPR.2017.549.

[99] L. Mou, Y. Hua, and X. Xiang Zhu, "A relation-augmented fully convolutional network for semantic segmentation in aerial scenes," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 12408–12417, doi: 10.1109/CVPR.2019.01270.

[100] M. D. Zeiler and R. Fergus, "Visualizing and understanding convolutional networks," in *Proc. 13th Eur. Conf. Comput. Vis.* (Lecture Notes in Computer Science), vol. 8689, D. J. Fleet, T. Pajdla, B. Schiele, and T. Tuytelaars, Eds. Zurich, Switzerland: Springer, 2014, pp. 818–833, doi: 10.1007/978-3-319-10590-1_53.

[101] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 25, 2012.

[102] M. Volpi and D. Tuia, "Dense semantic labeling of subdecimeter resolution images with convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 881–893, Feb. 2016, doi: 10.1109/TGRS.2016.2616585.

[103] D. Marcos, M. Volpi, B. Kellenberger, and D. Tuia, "Land cover mapping at very high resolution with rotation equivariant CNNs: Towards small yet accurate models," 2018, *arXiv:1803.06253*.

[104] G. Chen, X. Zhang, Q. Wang, F. Dai, Y. Gong, and K. Zhu, "Symmetrical dense-shortcut deep fully convolutional networks for semantic segmentation of very-high-resolution remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 5, pp. 1633–1644, May 2018, doi: 10.1109/JSTARS.2018.2810320.

[105] H. Luo, C. Chen, L. Fang, X. Zhu, and L. Lu, "High-resolution aerial images semantic segmentation using deep fully convolutional network with channel attention mechanism," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 9, pp. 3492–3507, Sep. 2019, doi: 10.1109/JSTARS.2019.2930724.

[106] Y. Liu, B. Fan, L. Wang, J. Bai, S. Xiang, and C. Pan, "Semantic labeling in very high resolution images via a self-cascaded convolutional neural network," 2018, *arXiv:1807.11236*.

[107] F. Zhang *et al.*, "ACFNet: Attentional class feature network for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Seoul, South Korea, Oct. 2019, pp. 6797–6806, doi: 10.1109/ICCV.2019.00690.

[108] Y. Sun, Y. Tian, and Y. Xu, "Problems of encoder-decoder frameworks for high-resolution remote sensing image segmentation: Structural stereotype and insufficient learning," *Neurocomputing*, vol. 330, pp. 297–304, Feb. 2019, doi: 10.1016/j.neucom.2018.11.051.

[109] N. Audebert, B. Le Saux, and S. Lefèvre, "Beyond RGB: Very high resolution urban remote sensing with multimodal deep networks," 2017, *arXiv:1711.08681*.

[110] Y. Liu, B. Fan, L. Wang, J. Bai, S. Xiang, and C. Pan, "Semantic labeling in very high resolution images via a self-cascaded convolutional neural network," 2018, *arXiv:1807.11236*.

[111] X. Li, Z. Zhong, J. Wu, Y. Yang, Z. Lin, and H. Liu, "Expectation-maximization attention networks for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Seoul, Korea (South, Oct. 2019, pp. 9166–9175, doi: 10.1109/ICCV.2019.00926.

[112] M. Yang, K. Yu, C. Zhang, Z. Li, and K. Yang, "DenseASPP for semantic segmentation in street scenes," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Salt Lake City, UT, USA, Jun. 2018, pp. 3684–3692. [Online]. Available: http://openaccess.thecvf.com/content_cvpr_2018/html/Yang_DenseASPP_for_Semantic_CVPR_2018_paper.html

[113] J. Fu *et al.*, "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, Jun. 2019, pp. 3146–3154. [Online]. Available: http://openaccess.thecvf.com/content_CVPR_2019/html/Fu_Dual_Attention_Network_for_Scene_Segmentation_CVPR_2019_paper.html

[114] H. Liu *et al.*, "An end-to-end network for panoptic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, Jun. 2019, pp. 6172–6181. [Online]. Available: http://openaccess.thecvf.com/content_CVPR_2019/html/Liu_An_End-To-End_Network_for_Panoptic_Segmentation_CVPR_2019_paper.html

[115] S. Zheng *et al.*, "Rethinking semantic segmentation from a sequence-to-sequence perspective with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 6881–6890. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2021/html/Zheng_Rethinking_Semantic_Segmentation_From_a_Sequence-to-Sequence_Perspective_With_Transformers_CVPR_2021_paper.html

[116] T. Xiao, Y. Liu, B. Zhou, Y. Jiang, and J. Sun, "Unified perceptual parsing for scene understanding," in *Proc. 15th Eur. Conf. Comput. Vis.* (Lecture Notes in Computer Science), vol. 11209, V. Ferrari, M. Hebert, C. Sminchisescu, and Y. Weiss, Eds. Munich, Germany: Springer, 2018, pp. 432–448, doi: 10.1007/978-3-030-01228-1_26.

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